



UAV control in autonomous object-goal navigation: a systematic literature review

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Abstract

Research interest in autonomous control of unmanned aerial vehicles (UAVs) has increased rapidly over the past decade. They are now widely used in civilian, military, and private areas. Applications include surveillance, search and rescue, and delivery tasks. More broadly, they excel at solving problems where a significant amount of space must be covered and traveled. However, using UAVs to solve navigation problems with full autonomy necessitates the mastering of complex subtasks. A solution that includes control, planning, localization, and mapping remains an open challenge. Object-goal navigation contains the same navigation problems where the main objective is to reach a target object. The search and identification of this target are central to the vehicle's navigation. This requires an understanding of what it is and where it can be located to move around the scene. This paper presents a systematic literature review on object-goal navigation and its subtasks, using autonomous UAVs. Survey taxonomies were found for the tasks and methods behind navigation and target localization problems using UAVs. The review analyzed 67 articles found between 2011 and 2022. They were found in the ACM, IEEE Xplore, WebOfScience, Science Direct, and Scopus databases. This review revealed essential issues related to autonomous navigation task dependencies. Moreover, it highlighted gaps in UAV development and framework standardization. Open challenges for autonomous UAV control for object-goal navigation must address the research on finding methods for problems. For example, autonomy level and comparison metrics, considering safety, ethics, and legal implications.

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1 Introduction

Unmanned aerial vehicles (UAVs) were first conceptualized at the beginning of the 20th century with the appearance of world conflicts (Mátyás and Máté 2019). When World War I began, flights were not only used for passenger delivery, and bombs were included. The first UAVs appeared that could fly to a target location, release the cargo, and return to base for reuse. Then, during World War II, UAVs were reduced in size, improving their distance range and control methods. In the Cold War, UAVs were used for remote sensing long-distance areas, taking pictures, and sensing and jamming radio frequencies. After these historical milestones, and until the end of the '90s, UAV technologies were improved with better and newer sensor ranges, new communication interfaces such as satellite links, and greater computational power.

In this century, many advances have improved UAV hardware technology, increasing their presence for civilian, educational, and commercial uses, among many others (Mohsan et al. 2022). An interesting research area is that focused on autonomous UAVs, where the implementation of the method is tightly constrained by their application, UAV type, and onboard sensors. For example, quadrotors are more efficient in agriculture due to their maneuverability and available visual sensors; nevertheless, carrying a payload becomes challenging (del Cerro et al. 2021). Another application for UAVs is in the Internet of Things, where communication extensions or computer power can be provided to the present devices (Liu et al. 2020).

UAVs present a significant advantage in many applications, improving a system's performance by performing tasks autonomously (Miranda et al. 2022). Additionally, the use of UAVs improves processes where a large area must be dealt with. A well-designed autonomous system must therefore allow a vehicle to move around safely and efficiently. First, an aerial vehicle flight over permitted areas without collision requires a navigation system. This system is responsible for offering suitable trajectories using sensory information. It allows the vehicle to move around the environment, ensuring travel from point A to point B. A movement strategy that considers travel speed and safety for the vehicle and the environment is important. Additionally, the vehicle must often carry out a given mission. In that case, the navigation system must also choose the point sequence that the vehicle must follow. Finally, a navigation system able to operate autonomously requires decision-making ability. It should be capable of determining suitable trajectories without human intervention.

The current literature has outlined different challenges in achieving autonomous navigation systems (ANS) (Yasuda et al. 2020). An ANS should solve problems such as control, localization, mapping, and path planning. Previous applications would occasionally include monitoring. As such, the sensing problem should also be addressed. In this regard and for simplicity, this document defines that an ANS must carry out four principal tasks: vehicle control, path planning, environment mapping, and surrounding perception.

Vehicle control is responsible for understanding how to apply force to move the vehicle at a desired velocity. For example, Song et al. (2022) uses a PID error feedback controller. The PID controller is named after its three component-equation that uses the output feedback to

measure an error value. Proportional-integral-derivative equations are applied to the error or difference between the current output and the desired set point. The PID controls the vehicle's direction along three axes, using the difference between current and desired location. This, enhanced with the Newton-Euler second-order derivative equations, can be used to control motor perturbation using position information.

Another challenge is path planning. This consists of evaluating the current vehicle state and defining where to go next. It proposes the next steps in a trajectory to reach its final location. Nitsche et al. (2020) outlines a two-stage method for path planning. The first stage creates a map using UAV orientation as a graph. In the second stage, a trajectory is created based on the map. This trajectory is proposed after comparing the current UAV location against the map. The proposed path encourages the UAV to follow the same trajectory presented on the map.

Path planning and mapping go hand in hand. Spatial representation is essential for planning, and the map is used to identify and locate objects within space. One of the objects can be the vehicle itself, as it is also part of the spatial representation. As such, localization problems are also considered mapping tasks. For this task, Kumar et al. (2020) proposed an occupancy grid map method. It used cells to divide an area, each with a corresponding probability of an object being present. Additionally, it implements Simultaneous Localization and Mapping (SLAM) to locate the UAV in a given map. SLAM is widely used to perform both tasks cost-efficiently. It provides a spatial representation of the environment and UAV attitude.

Finally, the perception task includes all the sensory data processing required by ANS methods. For example, Mumuni et al. (2022) proposed a vision-aided navigation system. It used three deep learning methods to process image depth, vehicle ego-motion, and camera optical flow. It is therefore capable of improving depth and object-level perception in dynamic, real-world environments. The resulting representation of the environment is used for control, obstacle avoidance, mapping, and planning. Most current papers address autonomous navigation as a point-driven or area-based problem. The vehicle must reach a defined coordinate point or area, respectively.

As mentioned above, an ANS handles the vehicle's trajectory from point A to point B. Object-goal navigation extends an ANS by seeking a target object while the vehicle travels around the scene (Batra et al. 2020). Therefore, identifying the target object is an additional problem beyond those presented in ANS. It requires object recognition and detection, which are necessary to identify the target's location. This kind of problem is commonly solved through computer vision (CV) techniques. However, the inclusion of CV methods in ANS mainly consists of adding the detection method as an independent process. Hence, the addition of CV increases the computational cost by processing visual sensor data solely for feature acquisition.

A notable challenge in object-goal navigation is the trade-off between navigation and exploration. Since all decisions should prioritize trajectories to areas with a high probability of finding the target object, the navigation behavior will travel around the known part of the scene. In contrast, if the known part does not contain the target object, the exploration behavior should select unknown locations with a high probability of finding it. In this regard, object-goal navigation methods should focus on the target's characteristics and where it could be found. Only a few papers implement UAVs for object-goal navigation, addressing this as an object search problem. This scarcity of UAV-based implementation for object-goal navigation is the main reason for conducting this systematic literature review (SLR).

This SLR seeks to identify the state-of-the-art object-goal navigation methods using UAVs for an emergency response scenario (Agrawal and Cleland-Huang 2021). Two main UAV types were found. The first is small and lightweight, able to fly faster with less noise. The other is large and capable of carrying a payload. A lightweight vehicle that can cover an area in a short time is desirable for monitoring emergencies. For example, in a fire monitoring task, the main objective is gathering visual information to identify the fire's location. Likewise, a larger vehicle is desirable for transporting objects that enable early assistance in a search and rescue task. Delivering medicines and/or food becomes essential if a person is found in a zone of difficult access. In both cases, the vehicle navigation is centered on finding the target object, a fire in the first example, and a person in the second.

This document also presents a well-defined methodology. It describes the search strategy, exclusion and inclusion criteria, information extraction, and quality criteria for article evaluation (Kitchenham and Charters 2007). This SLR attempts to classify, evaluate, and interpret the available research articles on object-goal navigation using UAVs. To the best of our knowledge, no previous works were found during the review's execution addressing object-goal navigation with UAVs. Only two surveys for UAV control regarding landing (Maitra et al. 2016) and obstacle avoidance (Sarmiento and Murphy 2018) were identified. Nevertheless, an example of visual navigation for mobile robots SLR (Yasuda et al. 2020) was found during manual searching. This task shares some aspects and diagnoses presented in this review. A main common issue is the lack of standardization in the experimental setup reported by the included papers.

The rest of the document is organized as follows: Section 2 introduces the methodology used for this systematic review. It is followed by the results in Section 3 with the taxonomies discovered through the research questions. Subsequently, Section 4 discusses the literature and open challenges, with the conclusions and final remarks outlined in Section 5.

2 Methodology

The research methodology used for the systematic literature review was based on the guidelines written by Kitchenham and Charters (2007). Its main purpose was to ensure reproducibility. Additionally, the SLR aimed to add scientific value by following a search strategy. This way, all relevant aspects of the research topic should be discovered. The review protocol itself was designed to find papers addressing object-goal navigation methods using aerial vehicles. First, the research question was defined. This guides the manuscript search, establishing the review's context and limitations. "Which characteristics and methods are required to solve an object-goal navigation problem using an unmanned aerial vehicle to explore, search, and recognize a target autonomously?" From this question, some specific research questions were also formulated and detailed in Table 1. The entire process is illustrated in Fig. 1. More details about the review protocol and execution can be found in Appendix A.

3 UAV-based autonomous object-goal navigation

No systematic literature review was found on UAV-based autonomous object-goal navigation between 2011 and 2022. A total of 67 articles included in this review were processed to extract

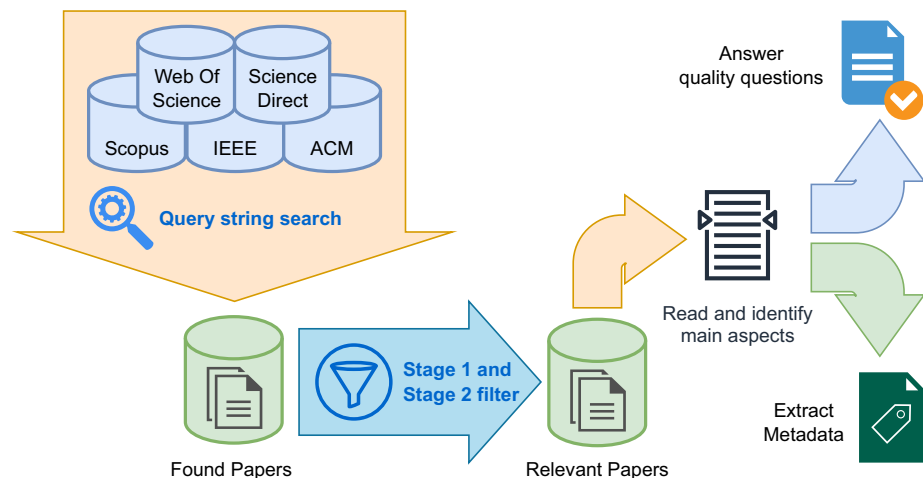


Fig. 1 Overall SLR method to obtain relevant papers and assess quality. On the left are the raw data, or papers obtained directly from the digital libraries. On the right are the selected papers, which were analyzed one by one. The results presented were obtained from quality questions and extracted metadata analysis of each paper

relevant data. Two of them were topic-related article surveys. The first is on bio-inspired methods for autonomous landing (Maitra et al. 2016). The second is on small UAVs' obstacle avoidance system (Sarmiento and Murphy 2018), inspired by animalistic flying behavior. Navigation-related papers and others focusing on navigation subproblems were found. The navigation-based papers try to automate the UAV's behavior, solving the necessary tasks to move from one given point to another. Other articles address autonomous tasks such as mapping, planning, controlling, and detection. Additionally, other tasks were found that complement the previous four, as detailed next.

The evolution of research interest in the topic was the first data to be processed. Figure 2 shows an increase in interest in recent years. Most of papers between 2020 and 2022 used a quadcopter vehicle type (38), followed by fixed-wing (19), multicopters (4), helicopters (3), and blimps (1). Some papers applied to more than one type of vehicle, as presented in Table 2.

The main difference between each UAV is based on its motion dynamics and thrust vector. Motion dynamics encompass all the principles that govern the motion of the vehicles, including the laws of physics. The vehicle's engine produces a thrust vector on the vehicle frame, which must be tilted to translate the vehicle. Initially, the thrust vector is parallel to the earth's surface for fixed-wing UAVs and blimps. However, the thrust vector is perpendicular to the earth's surface for rotor-propelled vehicles. As shown in Fig. 3,¹ each vehicle frame can rotate its orientation, illustrated by roll (ϕ), pitch (θ), and yaw (ψ) angles. The global frame is illustrated by E_x , E_y , and E_z , representing the three-dimensional axes.

The countries with the most research output on UAV-based autonomous object-goal navigation are China (21), the USA (13), and Germany (5). Other countries and their number of publications are presented in Fig. 6. Processing this metadata allows for a better understanding of interest in the research topic. Additionally, it introduces the context for a more

¹ Designed using assets from Freepik.com

Table 1 Research questions for the review

ID	Description	Goal
RQ1	Which tasks are involved in the object-goal navigation problem?	Identify the tasks or sub-problems defined by the authors.
RQ2	Which methods are used to obtain the desired goal searching behavior?	Identify the methods used for each task.
RQ3	What sensory vehicle information is used for each method in object-goal navigation?	Identify which data is used to solve each task.
RQ4	What metrics are used to benchmark the performance for object-goal navigation?	Identify how each proposal measures quality to compare it against others.
RQ5	What are the advantages and disadvantages of each method in object-goal navigation?	Identify utilities and drawbacks for all the proposals and room for improvements.

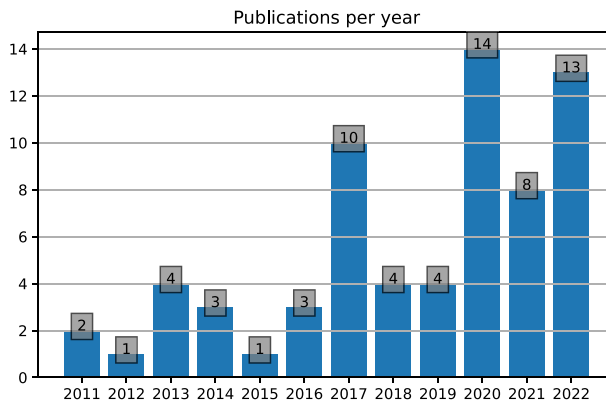


Fig. 2 Publications per year from the SLR

detailed analysis of the research questions. The following subsections present the findings of the processed data, classified into a grouping taxonomy for easier understanding.

3.1 Tasks involved

The first research question (RQ1) on the tasks involved in the object-goal navigation problems requires two primary behaviors, i.e., navigation and exploration. The main difference between these behaviors is the objective behind them. Navigation behavior was defined as the problem of displacing the vehicle from point A to point B in a fast and safe manner, while exploration behavior was defined as acquiring information from unvisited areas of the environment. As such, exploration must displace the vehicle to a more advantageous location. Only eleven articles included the described exploration behavior in their proposed method.

Four main tasks were defined within those two main behaviors. These are presented in Fig. 4:

- **Vehicle control** regulates the actuators' movement according to current pose and target coordinates. It requires real-time processing to ensure safety. The methods vary according to the vehicle's motion and degree of freedom.
- **Path planning** determines the series of points that the vehicle must reach. Succeeding coordinates establish a trajectory for exploration or navigation behaviors. The trajectory definition should ensure safe travel to the target location.
- **Scene mapping** represents the vehicle's surroundings, tagging free space, unexplored areas, and objects. This task is executed simultaneously with localization to identify the current vehicle's pose. Methods found use different sensorial information to create a map representation.
- **Environment perception** processes all sensor data for proper surrounding and vehicle descriptors. It is responsible for delivering insightful information for the other tasks. Proper environment details are essential to achieve the mission's goal.

The main tasks defined here matched those reported by Yasuda et al. (2020) in their SLR on visual navigation for mobile robots. The primary difference lies in the task of localization,

Table 2 Principal unmanned aerial vehicles types found in the literature

UAV type	Articles
Quadcopter: A four-propeller multirotor vehicle with a broader range of motion than a helicopter, advantageous for visual sensor use.	Abdulov et al. (2022); Bachrach et al. (2010); Bershadsky and Johnson (2013); Bhusal et al. (2022); Capi et al. (2021); Chen and Lu (2022); Cho et al. (2022); Cocchioni et al. (2014); Dang et al. (2018, 2020); Delamer et al. (2021); Fu et al. (2013, 2022); Fuentes et al. (2014); Graudo et al. (2020); Alarcon et al. (2021); Jia et al. (2022); Konrad et al. (2017); Kumar et al. (2020); Li et al. (2018); Mansouri et al. (2020b, a); Michaelsen et al. (2011); Nakamura et al. (2016, 2017); Narazaki et al. (2022); Nitsche et al. (2020); Parfiryev et al. (2020); Peake et al. (2020); Perez-Grau et al. (2021); Prophet and Trommer (2020); Sarmiento and Murphy (2018); Tsiourva and Papachristos (2020); Xu et al. (2013); Yang et al. (2021); Yu et al. (2017); Zhang et al. (2018); Zhao et al. (2021b)
Fixed-wing: Aerial vehicle that remains in constant motion following take-off, whose movement is constrained by thrust direction.	Cao (2015); He et al. (2017); Helgesen et al. (2019); Lee and Kim (2022); Li and Hu (2021); Louali et al. (2017); Lugo-Cárdenas et al. (2017); Meng et al. (2019); Mumuni et al. (2022); Nagarani et al. (2020); Sarmiento and Murphy (2018); Sun et al. (2019); Tymochko et al. (2020); Xu et al. (2013); You et al. (2020); Zhang et al. (2017); Zhao et al. (2021a); Zhu et al. (2018); Hu et al. (2022)
Helicopter: A multirotor vehicle with two propellers to control heading and motion to take off, land vertically, hover, and fly around.	Kusnur et al. (2021); Nakamura et al. (2017); de Plinval et al. (2017)
Multicopter: A multi-propeller vehicle, similar to a quadcopter, dragged by more than four motors.	Wang et al. (2017); Yousuf et al. (2022)
Blimp: A non-rigid vehicle without an internal structural framework. It moves by relying on lifting gas pressure and air-directed force.	Al-Jarrah et al. (2013)

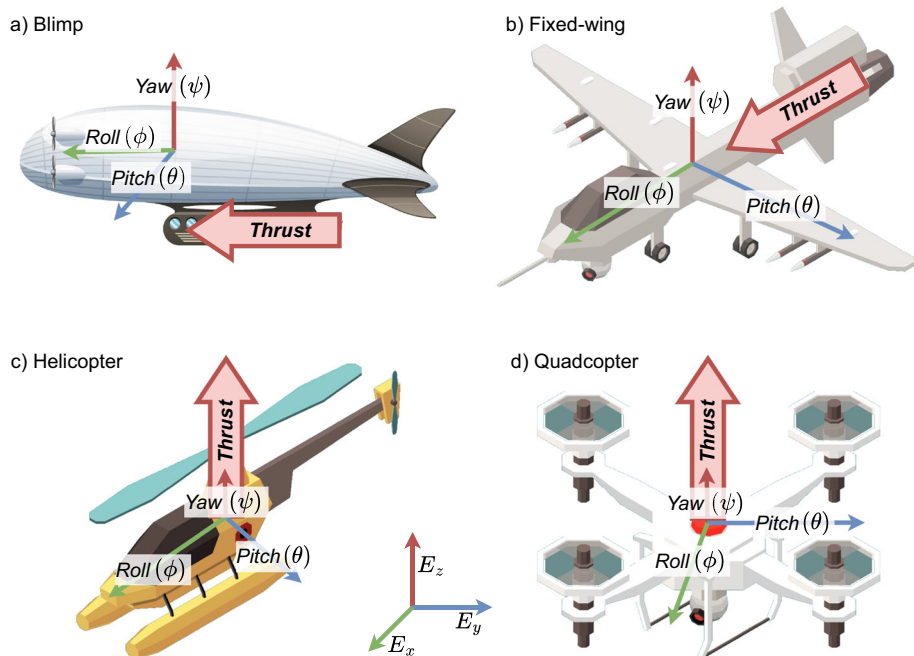


Fig. 3 Diagram of the principal unmanned aerial vehicle types found in the literature, with movement axes represented. E_x , E_y , E_z refer to the global frame. Roll (ϕ), pitch (θ), and yaw (ψ) represent the vehicle frame orientation. The red arrow thrust indicates each vehicle's initial force direction vector, with its changes allowing motion in the global frame

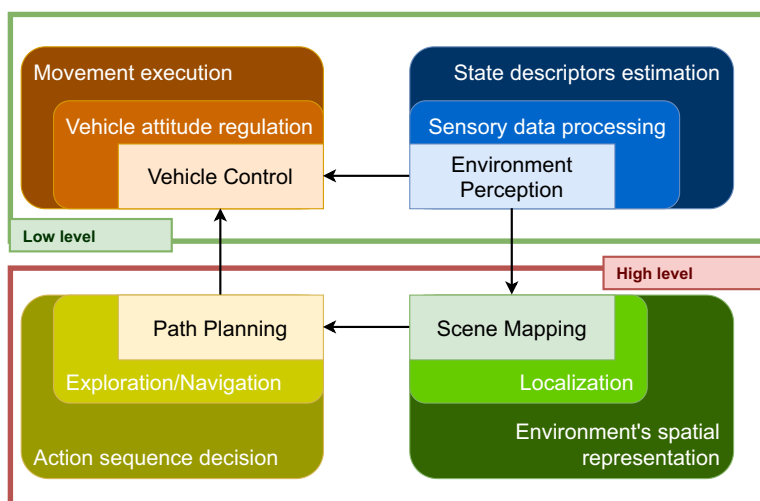


Fig. 4 The relationships between the main tasks required to achieve autonomous UAV navigation. The top row has the low-level tasks that must be executed in real-time for the UAV to operate safely in static or dynamic environments. The bottom row contains the high-level tasks that do not have to be processed in real-time. Nevertheless, they present information that can be insightful for the designed mission. The arrows indicate the dependencies of each task, e.g., the control task depends on perception and planning

which was included here as a mapping sub-task. A perception task was also included, which deals with all of the vehicle's sensor data.

Localization was included as a sub-task because, as presented by Peake et al. (2020), object-goal navigation can be solved using just the representation of the surrounding environment. This method was based on deep reinforcement learning and used sub-regions from an environment map. Results prove its efficiency compared to a hard-coded policy of navigating in zigzag.

Many applications perform localization and mapping simultaneously. SLAM presents the advantage of locating the UAV in the environment map, providing the vehicle's positional information. Localization then represents the vehicle's pose within the environment map. A UAV's localization can then be interpreted as the body frame coordinates and the map as the global frame coordinates. Both are representations of the environment from different perspectives. The purpose and detail level of the representation depends directly on the task's complexity. All the identified tasks are explained below.

As shown in Table 3, 50 papers present the control task in their proposal. A vehicle must influence its actuators or motors for navigation-only in 29 papers. In fixed-wing cases, the vehicle manipulates the wind to change the thrust direction. For rotor UAVs, different propeller velocities influences the thrust degree used to achieve movement. The presence of obstacles or structures can present a navigational challenge to vehicle displacement.

Specific AV control tasks such as landing and aerial refueling were found. Only two papers addressed aerial refueling where the UAV must approach a second AV. The approaching maneuvering must ensure the docking of a probe. Additionally, the UAV control must be steady and synchronized with the tanker. The tanker position and velocity are essential to ensure the refueling process. This task was considered only for fixed-wing UAVs.

The landing control task was found in nine papers, where the UAV is required to reach a ground position. In this task, a huge difference exists between rotor-propelled and fixed-wing UAVs. The first group only need a landing position because their thrust vector is vertical. The second group need a landing path for touchdown, reducing the horizontal thrust vector. Therefore, the vehicle's maneuvering should ensure the correct positioning in the landing path. Once positioned, it will reduce the thrust vector until it settles down.

Eleven papers deal with the control under exploration task. The control behavior must provide safe trajectories when in unknown environments. Therefore, vehicle control must reach undiscovered areas safely. The general purpose of exploration is to cover all available flight areas. Optionally, it can be used to search for interesting objects or areas.

The control tasks solved in four papers that address competition challenges rely upon other tasks. The perception or mapping tasks aid in locating the target's position. This is very helpful for landing and aerial refueling. Nevertheless, for exploration, planning tasks are essential.

Only 32 articles addressed path planning tasks, as shown in Table 4. Thirteen papers present a waypoint-based path planning. The vehicle must reach a series of predefined location points. A waypoint is commonly used when the map is known or when prior information is available. This planning relies on positional information to mark a route point as visited.

Thirteen articles do actual trajectory planning considering sensory inputs or a populated map. It then decides which location or coordinate the vehicle must reach. The main idea is to create a path to reach a target location. The trajectory chosen must be safe and as short as possible.

Seven articles deal with path planning for exploration purposes. Besides being safe, the chosen trajectory must balance unknown and known places. When the UAV revisits a previous place, it can improve the resolution or details of known objects. While going to unknown

Table 3 Control task scope found for autonomous object-goal navigation

Scope	Articles
Navigation: control for navigation applications	Al-Jarrah et al. (2013); Bhusal et al. (2022); Cao (2015); Capi et al. (2021); Chen and Lu (2022); Delamer et al. (2021); Fu et al. (2022); Grando et al. (2020); He et al. (2017); Helgesen et al. (2019); Jia et al. (2022); Kumar et al. (2020); Kusnur et al. (2021); Louali et al. (2017); Lugo-Cárdenas et al. (2017); Mansouri et al. (2020b); Nitsche et al. (2020); Parfiryev et al. (2020); Perez-Grau et al. (2021); de Plinval et al. (2017); Prophet and Trommer (2020); Song et al. (2022); Wang et al. (2017); You et al. (2020); Yousuf et al. (2022); Zhang et al. (2018); Zhao et al. (2021a, b); Hu et al. (2022)
Exploration: control for exploration application	Abdulov et al. (2022); Bachrach et al. (2010); Bershadsky and Johnson (2013); Dang et al. (2018, 2020); Fuentes et al. (2014); Alarcon et al. (2021); Mansouri et al. (2020a); Narazaki et al. (2022); Peake et al. (2020); Tsiourva and Papachristos (2020)
Landing: control applied on landing	Abdulov et al. (2022); Cho et al. (2022); Cocchioni et al. (2014); Li and Hu (2021); Meng et al. (2019); Nakamura et al. (2016, 2017); Xu et al. (2013); Yu et al. (2017)
Competition: control applied in competition	Abdulov et al. (2022); Bachrach et al. (2010); Bershadsky and Johnson (2013); Nakamura et al. (2016)
Refueling: control applied in aerial refueling	Sun et al. (2019); Zhu et al. (2018)

Table 4 Planning task scope found for autonomous object-goal navigation

Scope	Articles
Trajectory planning: path planning applications	Chen and Lu (2022); Delamer et al. (2021); Jia et al. (2022); Kumar et al. (2020); Kusnur et al. (2021); Lee and Kim (2022); Mao et al. (2020); Song et al. (2022); Tymochko et al. (2020); Yousuf et al. (2022); Zhang et al. (2017, 2018); Zhao et al. (2021b)
Waypoint: predefined points route applications	Cocchioni et al. (2014); Fu et al. (2022); Grando et al. (2020); He et al. (2017); Louali et al. (2017); Lugo-Cárdenas et al. (2017); Narazaki et al. (2022); Nitsche et al. (2020); Parfiriyev et al. (2020); de Plinval et al. (2017); Prophet and Trommer (2020); Wang et al. (2017); Yang et al. (2021)
Exploration: planning applied in exploration	Bachrach et al. (2010); Bershadsky and Johnson (2013); Dang et al. (2018, 2020); Alarcon et al. (2021); Narazaki et al. (2022); Tsiourva and Papachristos (2020)

places, it can detect objects not sensed before. Surrounding spatial information is therefore helpful for successful path planning.

Regarding the spatial tasks shown in Table 5, 34 papers addressed mapping or localization tasks. The mapping task creates a common spatial representation to provide surrounding information. The localization task identifies the vehicle pose within this spatial representation. A map representation requires global frame coordinates.

Only three articles presented a mapping-only task, representing coordinates for environment objects. That information is created in parallel with all other tasks, improving performance. Mapping constructs a spatial representation, with other tasks then querying that representation.

Similarly, sixteen papers deal with localization only. The main idea of localization is to infer relative coordinates from raw sensory data. Relative coordinates demand sensor landmarks to compare against, similar to direction finding. Sensed reference points enable the relative positioning of a given landmark.

Fifteen articles deal with mapping and localization simultaneously. The mapping task creates a representation of the surroundings, while the localization task determines the vehicle's pose within that representation. This not necessarily the same as the simultaneous localization and mapping method (SLAM). The SLAM method performs other processes to make both localization and mapping reliable. A good representation of the surroundings and vehicle localization depend on the perception task.

The perception task shown in Table 6 is essential for autonomous vehicle operations. Obstacles and targets are the two main objects that a UAV must recognize. Only 31 articles reported obstacle or target-aware methods. A two-class distinction between perception only and perception for control was found. Eight articles solve only the recognition of a given target or surrounding obstacles. The main challenge is on the target recognition side. An autonomous system needs specific descriptors to differentiate one object from another. Additionally, it requires accurate sensor data to produce these descriptors. The same challenge holds for obstacles, although descriptors can be more generic.

Twenty-three papers include the obstacles and target features in the control pipeline. Delivering insightful descriptors therefore becomes part of the vehicle state. Ideally, the perception task must deliver position or distance information, and deal with known and unknown tar-

Table 5 Mapping task scope found for autonomous object-goal navigation

Scope	Articles
Both: mapping and localization applications	Abdulov et al. (2022); Bachrach et al. (2010); Bershadsky and Johnson (2013); Dang et al. (2020); Delamer et al. (2021); Alarcon et al. (2021); Kumar et al. (2020); Kusunur et al. (2021); Li et al. (2018); Mao et al. (2020); Narazaki et al. (2022); Nitsche et al. (2020); Tsiourva and Papachristos (2020); Wang et al. (2017); Zhao et al. (2021b)
Localization: localization-only applications	Bhusal et al. (2022); Capi et al. (2021); Cho et al. (2022); Cocchioni et al. (2014); Dang et al. (2018); Helgesen et al. (2019); Konrad et al. (2017); Li and Hu (2021); Mansouri et al. (2020b); Meng et al. (2019); Perez-Grau et al. (2021); de Plinval et al. (2017); Sun et al. (2019); Xu et al. (2013); Yu et al. (2017); Zhu et al. (2018)
Mapping: mapping-only applications	Peake et al. (2020); Song et al. (2022); Yang et al. (2021)

Table 6 Perception task scope found for autonomous object-goal navigation

Scope	Articles
Control: obstacle & Target recognition applications with control	Abdulov et al. (2022); Al-Jarrah et al. (2013); Bachrach et al. (2010); Bershadsky and Johnson (2013); Capi et al. (2021); Chen and Lu (2022); Dang et al. (2020); Delamer et al. (2021); Fuentes et al. (2014); Grando et al. (2020); Alarcon et al. (2021); Kumar et al. (2020); Mansouri et al. (2020b, a); Nakamura et al. (2016); Narazaki et al. (2022); Peake et al. (2020); Perez-Grau et al. (2021); Prophet and Trommer (2020); Tsiourva and Papachristos (2020); You et al. (2020); Zhao et al. (2021b); Zhu et al. (2018)
Recognition: obstacle & Target recognition applications only	Albanese et al. (2022); Fu et al. (2013); Michaelson et al. (2011, 2012); Mumuni et al. (2022); Nagamani et al. (2020); Zaki et al. (2017); Zhang and Duan (2014)

gets and obstacles, in addition to environmental noise that can make it difficult to identify surrounding objects. Hence, the perception task requires a higher-order understanding of the environment.

The task relationship presented in Fig. 4 shows that perception is the most essential task. The perception task manages and processes all sensory data. Proper information must be delivered regardless of the environment. Aerial vehicle sensor data can range from one-dimensional to multi-dimensional. The more dimensions present, the more computationally expensive it becomes. Sometimes, better descriptors arise from multi-dimensional data. For example, visual sensors can determine the internal vehicle control state, objects in the scene, or the structure of the surroundings. Different papers address perception through diverse methods. However, most of them are probabilistic models that aim to obtain sensor information robust to noise. The richer the sensory data information flow, the better performance can be achieved.

3.2 Current methods

Answering RQ2, different methods were found which solve the tasks mentioned above. However, only a few articles addressed the navigation problem autonomously while others didn't address all of the necessary tasks. Additionally, some papers do not clarify what methods were used to solve some tasks. For example, in cases where the control method was omitted, the authors mention only the onboard control platform.

3.2.1 Control methods

Only 50 papers in Table 7 provided the implemented control method. 22 of them reported the use of autopilot-based control. Pixhawk autopilot is a vehicle control hardware platform that facilitates UAV development. It has easy-to-connect peripherals and mission management. Hardware-level control, executed in real-time, ensures the vehicle's safety and stability. Because the board is control-specific, most control tasks are solved at the hardware level. However, the board cannot process tasks that have a high computational demand. As such, this kind of solution is optimal for vehicle control only. All other high-level tasks must be implemented on another computer board.² Five papers use the PX4 software and the Pixhawk 4 advanced autopilot hardware. PX4 is a software application developed for vehicle operation, performing low-level control. It usually performs a vehicle's initial configuration and motor calibration. It can also execute more complex tasks such as mission planning.

Fourteen other articles used feedback error propagation methods, such as a Proportional-Integral-Derivative controller (PID controller) or Linear Quadratic Regulator (LQR). Two additional articles (de Plinval et al. 2017; Delamer et al. 2021) used a custom feedback controller implementation. Moreover, other articles include the PID controller with future value prediction through transition models (Tsiourva and Papachristos 2020; Dang et al. 2020), non-linear models (Mansouri et al. 2020b, a), and artificial neural networks (Fuentes et al., 2014). Generally, an error feedback method tries to minimize or maximize the control variable until it reaches a set point value. The difference between the input and the reference value is modified according to the implemented method. In the case of the PID, the proportional, integrative, and derivative mathematical methods are applied to calculate an output value. It is commonly to use two PID controllers to implement UAV control. One controller regulates

² https://docs.px4.io/main/en/getting_started/flight_controller_selection.html

Table 7 Control methods for autonomous object-goal navigation

Method	Articles
Autopilot: hardware-based control board	Abdulov et al. (2022); Bershadsky and Johnson (2013); Bhusal et al. (2022); Capi et al. (2021); Chen and Lu (2022); Cho et al. (2022); Cocchioni et al. (2014); Helgesen et al. (2019); Alarcon et al. (2021); Jia et al. (2022); Kusnur et al. (2021); Li and Hu (2021); Lugo-Cárdenas et al. (2017); Nakamura et al. (2017); Narazaki et al. (2022); Perez-Grau et al. (2021); Wang et al. (2017); Xu et al. (2013); Yu et al. (2017); Zhang et al. (2018); Zhao et al. (2021b); Zhu et al. (2018)
Error feedback: error-based correction methods	Bachrach et al. (2010); Delamer et al. (2021); Fuentes et al. (2014); Kumar et al. (2020); Louali et al. (2017); Mansouri et al. (2020b, a); Meng et al. (2019); Nakamura et al. (2016); Nitsche et al. (2020); Parfiryev et al. (2020); de Plinval et al. (2017); Song et al. (2022); Sun et al. (2019)
Learning algorithm: acquire a motion model	Al-Jarrah et al. (2013); Fu et al. (2022); Grando et al. (2020); He et al. (2017); Peake et al. (2020); You et al. (2020); Zhao et al. (2021a); Hu et al. (2022)
Probabilistic: future state inference	Dang et al. (2020); He et al. (2017); Helgesen et al. (2019); Lugo-Cárdenas et al. (2017); Mansouri et al. (2020b, a); Tsiourva and Papachristos (2020); Yousuf et al. (2022)
Others: custom controlling	Cao (2015); Dang et al. (2018); Prophet and Trommer (2020)

movement along the x-axis, and the other, the y-axis. The controller's output is always used as motor velocity. According to the error feedback method, the operations will be applied to the difference. These kinds of controllers are widely used and their hyperparameters are easily tuned. However, more than one controller is required to achieve smooth flight behaviors, thereby, increasing linearly the number of hyperparameters that must also be tuned.

The other eight papers address the control task with methods such as reinforcement learning (Fu et al. 2022; Zhao et al. 2021b) and fuzzy algorithms (Al-Jarrah et al. 2013). These methods aim to develop a suitable control policy to move the vehicle around the environment. These learning methods try to map the environment observation into a desired control output. The control behavior achieved by the UAV depends directly on the cost function used. In the case of reinforcement learning, the cost function is known as the reward function, a value that models the problem's goal. The policy obtained will depend directly on the control problem modeling. Most policies learn to go from one point to another as an end-to-end solution. Applying learning methods for UAV control has demonstrated that high-level commands can be used to achieve movement. However, the main issue is the number of samples required to solve the problem. Most of the papers found propose methods to reduce learning time and improve performance.

Eight articles present a predictive or probabilistic method for control tasks. These methods process the vehicle's state to aid the vehicle control algorithm. Yousuf et al. (2022) implement a probability of density hypothesis to filter possible objects' locations. Tsiourva and Papachristos (2020) propose a visual-thermal saliency model to enhance control inside an underground mine. Mansouri et al. (2020b) present a state estimator to infer the next state, constraining control inside an underground mine. Interesting dual control cycle method implementations were found in Helgesen et al. (2019) and Lugo-Cárdenas et al. (2017). The first is focused on vehicle stability and attitude control, and the second manages high-level movements. Both methods control the UAV's behavior to reach a defined target point. They also improve vehicle movement, providing a smooth and safe trajectory. However, the application is constrained to specific environments, presenting a low generalization.

3.2.2 Planning methods

The planning methods found from Table 8 differ with regard to the objective of each behavior. Twelve papers address the planning method as a waypoint following problem. In this case, the vehicle must correct its pose to reach the next waypoint in the route. Most waypoint-following papers present a navigation-only behavior.

Ten papers focused on the planning method as a graph-based approach. They use a cost function over the graph's vertices to determine which point to reach. Among the graph-based methods, several tree-search algorithms were found. They were Rapidly-exploring Random Tree (RRT) (Dang et al. 2018; Prophet and Trommer 2020), its improved version RRT* (Alarcon et al. 2021), the Dijkstra algorithm (Bershatsky and Johnson 2013), and frontier-based exploration (Bachrach et al. 2010). The main idea behind graph-based methods is to go from one node to another, traveling the shortest obstacle-free path based on vertex cost analysis between the required nodes.

Other methods are proposed that use genetic (Tymochko et al. 2020; Song et al. 2022) or optimization (Delamer et al. 2021; Jia et al. 2022) algorithms. An interesting approach was presented by Zhao et al. (2021b), with intermediate goals to reach the final known point. The methods in Others category, includes custom implementations and mathematical models.

In summary, all planning methods uses UAV position and the problem objective to plan tasks executions such as position, velocity, or future state that must be reached. The planning

Table 8 Planning methods for autonomous object-goal navigation

Method	Articles
Waypoint following; predefined trajectory	Cocchioni et al. (2014); Grando et al. (2020); He et al. (2017); Kumar et al. (2020); Louali et al. (2017); Lugo-Cárdenas et al. (2017); Nitsche et al. (2020); Parfiryev et al. (2020); de Plinval et al. (2017); Prophet and Trommer (2020); Wang et al. (2017); Yang et al. (2021)
Graph-based: edge and node analysis	Abdulov et al. (2022); Bachrach et al. (2010); Bershadsky and Johnson (2013); Dang et al. (2018, 2020); Alarcon et al. (2021); Kusnur et al. (2021); Mao et al. (2020); Prophet and Trommer (2020); Tsiourva and Papachristos (2020)
Algorithms: optimization methods	Jia et al. (2022); Song et al. (2022); Tymochko et al. (2020); Zhang et al. (2018)
Others: custom planning	Chen and Lu (2022); Delamer et al. (2021); Fu et al. (2022); Jia et al. (2022); Kumar et al. (2020); Lee and Kim (2022); Narazaki et al. (2022); Yousuf et al. (2022); Zhang et al. (2017); Zhao et al. (2021b)

methods are focused on optimizing a score value given a function or constraints, to find the shortest path to a target location. Most of the planning implementations uses information such as UAV position, pose, and velocity, to reach a target state. However, when the target state includes the object's location, a planning method must choose or infer where the object could be.

3.2.3 Mapping methods

Most planning methods require prior spatial information to propose a movement scheme. The mapping task solves the spatial information problem through three main types of methods. As shown in Table 9, ten papers treat the mapping task as reconstruction. They fuse sensory data to create a spatial projection for each element in the scene. Simultaneous mapping and localization (SLAM) (Bachrach et al. 2010; Bershadsky and Johnson 2013; Dang et al. 2020; Zhao et al. 2021b), a widely used navigation method, solves this reconstruction problem. It fuses internal and external sensor information to create a map with structural information such as corridors, objects, and the vehicle itself. Another method is called iterative closest point (ICP) (Li et al. 2018; Narazaki et al. 2022), which aims to improve approaches that connect a point cloud to an image obtained from sensors. This relation adds visual depth information to the map. Other map reconstruction methods are Quadtree decomposition (Peake et al. 2020), Oblique photogrammetry (Yang et al. 2021), and laser sensor based implementation (Tsiourva and Papachristos 2020). These methods, like ICP, aim to fuse visual and depth sensor data to create a map that includes distance information about the surrounding environment structure and the objects within it.

Another mapping task category was defined as representation, with seven papers employing this method. The major difference encountered between the representation and reconstruction methods introduced earlier is how they use sensor data. Reconstruction creates a map that includes all surrounding elements, while representation uses that mapped information to form a spatial projection of the environment. In simple terms, the representation of a map is simpler and contains only the information vital to the navigation problem. The map representation methods commonly use occupancy grid maps, which are separated into different cell scene regions. A variation of occupancy maps adds cell priority (Kusnur et al. 2021) according to a computed score or adds OctoMap (Dang et al. 2018), which includes the Octree data structure for efficiency. Other variations were implemented with SLAM for sensory data fusing (Dang et al. 2020; Zhao et al. 2021b).

A final category for the mapping task is the reference map. These methods use environment information as a reference for spatial information. This mapping approach lets a vehicle know its position after sensing the reference information. Only two methods were found with this kind of mapping. The first (Nitsche et al. 2020) used visual landmarks to locate an interesting spatial region in the scene. The second used a radio frequency (RF) signal (Wang et al. 2017) to find objects of interest, creating a signal fingerprint of the distance between the vehicle and different objects' beacons in the environment.

3.2.4 Localization methods

Many mapping methods, such as SLAM-based ones, require the current vehicle pose to create correct spatial information that considers distance and coordinates. However, localization relies mainly on odometry sensor data and vehicle control state to estimate the current vehicle information. Four main localization categories were found in the literature review, which are presented in Table 10.

Table 9 Mapping methods for autonomous object-goal navigation

Method	Articles
Reconstruction: replica from sensor data	Bachrach et al. (2010); Bershadsky and Johnson (2013); Dang et al. (2020); Alarcon et al. (2021); Li et al. (2018); Narazaki et al. (2022); Peake et al. (2020); Tsiourva and Papachristos (2020); Yang et al. (2021); Zhao et al. (2021b)
Reference: environment marks measurement	Nitsche et al. (2020); Wang et al. (2017)
Representation: environment projection into another space	Abdulov et al. (2022); Dang et al. (2018, 2020); Delamer et al. (2021); Kumar et al. (2020); Kusnur et al. (2021); Mao et al. (2020); Song et al. (2022); Zhao et al. (2021b)

Table 10 Localization methods for autonomous object-goal navigation

Method	Articles
Autopilot: hardware-based sensor communication	Alarcon et al. (2021)
Probabilistic: sensor data modelling	Abdulov et al. (2022); Bachrach et al. (2010); Bershadsky and Johnson (2013); Cocchioni et al. (2014); Dang et al. (2020); Delamer et al. (2021); Helgesen et al. (2019); Konrad et al. (2017); Kumar et al. (2020); Kusnur et al. (2021); Li et al. (2018); Li and Hu (2021); Mansouri et al. (2020b); Meng et al. (2019); Narazaki et al. (2022); Wang et al. (2017); Zhao et al. (2021b)
Reference: landmark recognition	Capi et al. (2021); Cho et al. (2022); Perez-Grau et al. (2021); de Plinval et al. (2017); Sun et al. (2019); Xu et al. (2013); Yu et al. (2017); Zhu et al. (2018)
Visual-based: camera-based estimations	Abdulov et al. (2022); Bhusal et al. (2022); Dang et al. (2018); Mao et al. (2020); Nitsche et al. (2020); Tsiourva and Papachristos (2020); Xu et al. (2013)

One category called probabilistic, had seventeen articles using this method. They process the vehicle sensor data with probability techniques to infer an expected state. As the SLAM technique processes sensor data to correct the current vehicle pose (Bachrach et al. 2010; Bershadsky and Johnson 2013; Cocchioni et al. 2014; Dang et al. 2020; Narazaki et al. 2022; Kumar et al. 2020; Zhao et al. 2021b), was included in this category. Additionally to SLAM, some methods complemented it with probability methods such as Kalman filters. There can also be variations such as extended Kalman filters (EKF) (Konrad et al. 2017; Li and Hu 2021; Mansouri et al. 2020b; Meng et al. 2019) and Kalman consensus filter (Li et al. 2018), non-linear observer (Helgesen et al. 2019), particle filters (Wang et al. 2017), and Monte Carlo processes (Delamer et al. 2021; Kusnur et al. 2021). The probabilistic method's central idea is to infer the control vehicle's state for accurate pose interpretation. This inference is optimized during execution, constantly updating over time to be noise-robust.

Another category was named reference localization. This is where the vehicle pose is estimated using signal strength sensor data to define a target pose. In this category, eight articles were found. Three of them used visual references for state estimation, to reach a target state for aerial refueling (Sun et al. 2019; Zhu et al. 2018; Lee and Kim 2022). Other methods used a matrix barcode (Yu et al. 2017) observed by a camera to estimate pose. The 2D barcode was created to represent an aerial vehicle's six degrees of freedom. Similarly, two other papers use 2D ARuco markers as a visual reference to identify the target direction (Capi et al. 2021) and the landing pad position (Cho et al. 2022). Another paper presented a radio frequency-based method (Perez-Grau et al. 2021) to infer current vehicle position by processing each object's signal strength. Each signal presents a custom identification, allowing differentiation of one beacon signal from another for correct position triangulation. Custom localization proposals were also found (Xu et al. 2013), integrating thermal images to locate points of interest used as a reference for autonomous landing. Another article presented a pose target as a reference (de Plinval et al. 2017) to obtain the current relative vehicle pose and reach the objective point.

The third localization method category was that of visual-based approaches. They process image data with or without additional odometry information to estimate current vehicle pose. Seven articles were included in this category. One custom proposal (Mao et al. 2020) uses landmark recognition and a confidence score to know the current position while traveling from one point to another. The other three articles fuse ?? the visual and inertial information to achieve accurate state estimation. This latter method can track the vehicle's position under motion without a map or any prior information. A final category with only one paper (Alarcon et al. 2021) relies on the autopilot system, which delegates the localization method to the onboard system.

3.2.5 Perception methods

The perception methods identified during the review were classified into four categories in Table 11. A category with nineteen papers is computer vision (CV) based approaches. CV methods use hand-crafted features to interpret sensor data, mainly from visual sensors. Hand-crafted features or descriptors were captured after being processed by different CV techniques. Descriptors highlight aspects such as color, border, textures, intensity changes, and motion. Nakamura et al. (2016) present a CV method with EKF to improve the accuracy of state estimations under a noise-robust setup. A saliency map was also used with CV techniques by Tsiourva and Papachristos (2020), highlighting the zone of interest from the visual field. Narazaki et al. (2022) proposed a Structure from Motion (SfM) method with computer vision to capture geometrical information on the surroundings.

Table 11 Perception methods for autonomous object-goal navigation

Method	Articles
Computer vision: manual descriptors selection	Abdulov et al. (2022); Al-Jarrah et al. (2013); Bershadsky and Johnson (2013); Bhusal et al. (2022); Capi et al. (2021); Cocchioni et al. (2014); Fuentes et al. (2014); Helgesen et al. (2019); Kumar et al. (2020); Li et al. (2018); Michaelsen et al. (2011, 2012); Nagarani et al. (2020); Nakamura et al. (2016); Narazaki et al. (2022); Tsiourva and Papachristos (2020); Xu et al. (2013); Yu et al. (2017); Zhang and Duan (2014)
Probabilistic: current state estimations	Bachrach et al. (2010); Chen and Lu (2022); Cho et al. (2022); Delamer et al. (2021); Mansouri et al. (2020a); Meng et al. (2019); Nakamura et al. (2016, 2017); Perez-Grau et al. (2021); Zhu et al. (2018)
Learning-based: acquire useful state descriptor	Abdulov et al. (2022); Albanese et al. (2022); Bhusal et al. (2022); Dang et al. (2018, 2020); Fu et al. (2022); Grando et al. (2020); Li and Hu (2021); Mansouri et al. (2020b); Mumuni et al. (2022); Peake et al. (2020); You et al. (2020); Zaki et al. (2017)
Other: custom sensor process	Chen and Lu (2022); Fu et al. (2013); Prophet and Trommer (2020); Zhao et al. (2021b)

More elaborate systems require more than one sensor processing method. For example, Bhusal et al. (2022) use CV to locate the 3D position of a flock of birds after detecting movement using convolutional neural networks (CNN). Abdulov et al. (2022) use CV methods in simple mark recognition and object detection algorithms to obtain the coordinates of specific objects. In this category, Zhang and Duan (2014) proposed a target detection method through template matching. Template matching relies on prior knowledge of the target's features to compare against visual sensor information, aiming to locate the target correctly within the image. In this regard, Fuentes et al. (2014) proposed an entropy-based value to check how many possible interest objects are in the scene. Furthermore, they processed the image with computer vision methods to identify landmarks.

Ten articles were classified as using probabilistic methods to process sensory information. Techniques such as EKF obtain a proper state estimation from current captured sensor data. Nakamura et al. (2017) uses EKF with a particle filter to achieve correct landing control.

Another category includes thirteen papers with learning methods based on the gradient-descent algorithm. These methods use artificial neural networks (ANN) to transform sensor data into another latent representation. Neural networks can learn noise robust sensor representation depending on the training process. Two main ANN kinds were found, shallow fully-connected networks and convolutional neural networks (CNN). Both can process any numerical input data, however, the CNN are widely used to capture features from higher dimensional data. From learning methods, Li and Hu (2021) implement two-hierarchy CNN to achieve autonomous landing from a ground-based vision sensor. Mansouri et al. (2020b) include geometric vector representation in the CNN processing pipeline to acquire useful sensor information on the surroundings. Other methods use the YOLO detector to locate and identify different surrounding elements. Dang et al. (2018) and Alarcon et al. (2021) used YOLO (Redmon et al. 2016) mainly to identify the target object under autonomous navigation. However, in Dang et al. (2020), the detector was aided with bayesian filters to improve the object identification and depth-related information.

A final category of custom implementations includes four papers. One uses a tracking-only method (Fu et al. 2013) to report the target's position from visual input. Other custom implementations use Density-Based Spatial Clustering of Applications with Noise (DBSCAN) approach (Chen and Lu 2022; Prophet and Trommer 2020) to produce surrounding information from laser data. Another method processes the sensor data as spatial-structure information (Zhao et al. 2021b) to solve the autonomous navigation problem.

3.3 Sensory vehicle information

During the review it was discovered that different authors use different sensor data for the control, perception, mapping, and planning tasks. Nevertheless, it was possible to identify that most methods use visual sensors, followed by reference, range, and position sensors. The visual sensors use any visible or invisible light spectra. The control task is the one in which more sensor types were reported, followed by localization, planning, mapping, perception (obstacle and target recognition), and exploration.

3.3.1 Sensors classification

One first sensor category presented in Table 12 was named visual sensors. 42 articles use image signals as primary input data for their proposals. Most use RGB images, in a monocular setup. Some papers combined RGB with depth image information (Chen and Lu 2022; Li

Table 12 Visual sensors classification for autonomous object-goal navigation

Sensor	Articles
Monocular RGB image: single camera	Al-Jarrah et al. (2013); Bachrach et al. (2010); Bershadsky and Johnson (2013); Bhusal et al. (2022); Capi et al. (2021); Chen and Lu (2022); Cho et al. (2022); Cocchioni et al. (2014); Dang et al. (2020); Fu et al. (2013, 2022); Fuentes et al. (2014); Kumar et al. (2020); Kusnur et al. (2021); Li et al. (2018); Mansouri et al. (2020b, a); Mao et al. (2020); Michaelsen et al. (2011, 2012); Nagarani et al. (2020); Nakamura et al. (2016, 2017); Narazaki et al. (2022); Peake et al. (2020); de Plinval et al. (2017); Prophet and Trommer (2020); Song et al. (2022); Sun et al. (2019); Yu et al. (2017); Zaki et al. (2017); Zhao et al. (2021b)
Stereo RGB image: dual cameras	Abdulov et al. (2022); Bhusal et al. (2022); Dang et al. (2018); Alarcon et al. (2021); Li and Hu (2021); Nitsche et al. (2020); Tsiourva and Papachristos (2020); Yang et al. (2021)
Depth image: distance camera	Chen and Lu (2022); Alarcon et al. (2021); Li et al. (2018); Zaki et al. (2017); Zhao et al. (2021b)
Thermal image: infrared cameras	Cocchioni et al. (2014); Dang et al. (2020); Helgesen et al. (2019); Meng et al. (2019); Tsiourva and Papachristos (2020); Xu et al. (2013)

et al. 2018; Zaki et al. 2017; Zhao et al. 2021b), or used RGB with thermal image information (Cocchioni et al. 2014; Dang et al. 2020). Others use a stereo vision setup combined with depth (Alarcon et al. 2021) or thermal images (Tsiourva and Papachristos 2020). Finally, other papers use only thermal images (Meng et al. 2019; Xu et al. 2013; Helgesen et al. 2019). These visual sensors were used mainly to identify landmarks, objects, or points of interest around the vehicle.

Table 13 presents the second sensor category, data called reference. 34 articles used sensor data that depends on a reference framework to obtain the different measurements. Almost all papers included used inertial measurement units (IMU). These are sensor information that measures and reports a body's specific force. However, Capi et al. (2021) used an accelerometer and magnetometer, while Bhusal et al. (2022) used a barometer only. An IMU presents a combination of accelerometer and gyroscope sensors from an aerial vehicle's six degrees of freedom. In this category, only Meng et al. (2019) and Nakamura et al. (2016) complement the IMU sensor with an altimeter or magnetometer. This improves localization in autonomous landings. The altimeter measures the vehicle's altitude in relation to sea level using atmospheric pressure as a reference. Likewise, the barometer measures the air pressure in a particular environment. The magnetometer uses the north pole as a reference, measuring the vehicle's heading angle offset from north. The reference sensors are important for the vehicle's attitude control, allowing stability, localization information, and next position estimations.

A third category is that of range sensors, which are mentioned in 22 articles, presented in Table 14. Range sensors measure distance, assessing the time required for an emitted signal to return. Most articles in this category present the Light Detection and Ranging (LiDAR) sensor. The LiDAR sensor uses laser technology to measure the time the reflected light takes to return to the receiver. Other articles use Sound Navigation and Ranging (Sonar), which use sound propagation to measure distances. Sonar technology is commonly used in maritime vehicles such as submarines. Bershadsky and Johnson (2013) and Kumar et al. (2020) combine LiDAR and Sonar sensors to achieve autonomous navigation. He et al. (2017); You et al. (2020); Hu et al. (2022) use Radio detection and ranging (Radar), where radio waves determine the distance, angle, and radial velocity. These kinds of sensors were used in two main cases. The first was to identify solid objects such as walls or obstacles in the surroundings, aiming to avoid them. The second was to add or improve depth measurement from visual input. Range sensors and visual input fusion aid in accurately locating the object in the scene and projecting into a new spatial representation.

Finally, a fourth category represents the position sensors, as shown in Table 15. In this category, ten papers presented a sensor based on geo-positional information. Most depend on the Global Position System (GPS) to obtain the vehicle's current position. Other papers (Delamer et al. 2021; Konrad et al. 2017) did not specify which positional system was used, defining it only as a Global Navigation Satellite System (GNSS) sensor. A GPS sensor is a specific type of GNSS sensor that relies on a satellite constellation to measure position in latitude and longitude. This kind of sensor depends on satellite availability and signal strength, requiring more than one satellite to work.

3.3.2 Vehicle control state

Some articles reported vehicle state information using the sensor data mentioned above. These are presented in Table 16. In 34 of these articles there were no details regarding the state, or it was impossible to determine. Another fourteen papers did not require state information

Table 13 Reference sensors classification for autonomous object-goal navigation

Sensor	Articles
Inertial movement unit: force and angle rates	Bachrach et al. (2010); Bershadsky and Johnson (2013); Cao (2015); Chen and Lu (2022); Cocchiomi et al. (2014); Dang et al. (2018, 2020); Delamer et al. (2021); Grando et al. (2020); He et al. (2017); Helgesen et al. (2019); Konrad et al. (2017); Kumar et al. (2020); Louali et al. (2017); Lugo-Cárdenas et al. (2017); Mansouri et al. (2020b, a); Meng et al. (2019); Mumuni et al. (2022); Nakamura et al. (2016, 2017); Nitsche et al. (2020); Parfiriev et al. (2020); Perez-Grau et al. (2021); de Plinval et al. (2017); Prophet and Trommer (2020); Sun et al. (2019); Tsiourva and Papachristos (2020); Wang et al. (2017); Xu et al. (2013); Zhao et al. (2021b); Zhu et al. (2018)
Magnetometer: North heading	Capi et al. (2021); Nakamura et al. (2016)
Altimeter and Barometer: atmospheric pressure	Bhusal et al. (2022); Meng et al. (2019)
Accelerometer: angular acceleration	Capi et al. (2021)
Gyroscope: angular velocity	Hu et al. (2022)

Table 14 Range sensors classification for autonomous object-goal navigation

Sensor	Articles
LiDAR: light ranging	Abdulov et al. (2022); Bachrach et al. (2010); Bershadsky and Johnson (2013); Chen and Lu (2022); Dang et al. (2020); Grando et al. (2020); Alarcon et al. (2021); Kumar et al. (2020); Mansouri et al. (2020b, a); Narazaki et al. (2022); Perez-Grau et al. (2021); Prophet and Trommer (2020); Song et al. (2022); Tsiourva and Papachristos (2020); Wang et al. (2017); Zhu et al. (2018)
	Bershadsky and Johnson (2013); Cocchioni et al. (2014); Kumar et al. (2020); Nakamura et al. (2016)
Sonar: sound ranging	He et al. (2017); You et al. (2020); Hu et al. (2022)
Radar: radio signal ranging	

Table 15 Position sensors classification for autonomous object-goal navigation

Sensor	Articles
GPS: global position system	Cho et al. (2022); Cocchioni et al. (2014); Helgesen et al. (2019); Louali et al. (2017); Narazaki et al. (2022); Sun et al. (2019); Zhu et al. (2018); Hu et al. (2022)
GNSS: global navigation satellite system	Delamer et al. (2021); Konrad et al. (2017)

Table 16 Vehicle control state information used for autonomous object-goal navigation

State data	Articles
Position: Coordinates in a map reference system	Bachrach et al. (2010); Cao (2015); Cocchioni et al. (2014); He et al. (2017); Konrad et al. (2017); Kumar et al. (2020); Kusnur et al. (2021); Louali et al. (2017); Lugo-Cárdenas et al. (2017); Meng et al. (2019); Nakamura et al. (2016, 2017); Song et al. (2022); You et al. (2020); Yu et al. (2017); Zhu et al. (2018); Hu et al. (2022)
Orientation: heading angle	Al-Jarrah et al. (2013); Bachrach et al. (2010); Cao (2015); He et al. (2017); Konrad et al. (2017); Kumar et al. (2020); Lee and Kim (2022); Louali et al. (2017); Lugo-Cárdenas et al. (2017); Meng et al. (2019); Nitsche et al. (2020); Song et al. (2022); Yu et al. (2017); Zhang et al. (2017)
Movement: acceleration or velocity	Bachrach et al. (2010); Grando et al. (2020); Konrad et al. (2017); Kusnur et al. (2021); Mansouri et al. (2020b, a); Meng et al. (2019); Nakamura et al. (2016, 2017); Nitsche et al. (2020); Song et al. (2022); Wang et al. (2017); You et al. (2020); Zhang et al. (2017); Zhu et al. (2018); Hu et al. (2022)
Bias: Central reference data value	Bachrach et al. (2010); Konrad et al. (2017); Nakamura et al. (2016); Nitsche et al. (2020)
Angle: angulation degree of freedom	Fuentes et al. (2014); Grando et al. (2020); Kusnur et al. (2021); Meng et al. (2019); Zhao et al. (2021a); Hu et al. (2022)
Target: target's relative information	Al-Jarrah et al. (2013); Grando et al. (2020); Lee and Kim (2022); Peake et al. (2020); Zhao et al. (2021a)

because they presented a no-control proposal. Only 30 articles successfully described the vehicle state.

Seventeen articles include the vehicle position as three or two-axis coordinates in an absolute system or calculated from a reference system. Another eight papers considered the use of vehicle orientation as essential, indicating the vehicle pose with a reference system. Movement information was found in fourteen papers, where the authors used the velocity, acceleration, or attitude data in the control state. Additional relative information was reported in ten papers with data such as gyroscope, acceleration, or heading angle. This relative information was measured in relation to a central defined value. Furthermore, five articles used information relative to the target, such as distance or direction.

Grando et al. (2020)'s paper uses laser rays to achieve a collision-free proposal. In Li and Hu (2021), the authors compute the 3D coordinates of the aerial vehicle from ground cameras. For the proposal of Delamer et al. (2021), the problem was defined as a Markov decision process with two types of states. The first state is fully-observable and obtained from sensory data. The second is partially-observable and obtained from the vehicle's attitude estimation. For Peake et al. (2020), the state's definition is obtained according to the vehicle's behavior. Both the target search and region exploration behaviors use probabilistic target detection. Additionally, there is an interesting use of the gravity direction in the state definition presented by Nitsche et al. (2020). Finally, Kusnur et al. (2021) include the timestamp as a state value.

3.4 Comparison benchmarks

For research question RQ4, distinct benchmark metrics and environment tests were found. Most papers performed a quantitative evaluation. One kind of benchmark metric found used an error rate measure. The error calculation methods found were root mean square (RMS), mean average error (MAE), and relative pose error (RPE). A few articles presented a custom error calculation. For example, Zhang et al. (2018) present a custom cost function, used in an optimization algorithm as a quality metric. Mao et al. (2020) proposed a bio-inspired method that compares the path integration value against a proposed path. They used MAE to differentiate the proposed trajectory from the one proposed by the method in three different cases. The MAE can be formalized as follows,

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |e_i|,$$

with $|e_i| = |y_i - x_i|$, where y_i and x_i are the predicted and true values for n samples.

He et al. (2017) used the RMS to compare the range estimation error within a given time window. The estimation error was obtained from the difference between a baseline path and the one inferred by the methods. The RMS can be formalized as follows,

$$\text{RMS} = \sqrt{\frac{1}{n} \sum_{i=1}^n e_i^2},$$

with n as the number of errors, and e the error value.

The Nitsche et al. (2020) proposal uses the RPE metric to evaluate a visual odometry method. This evaluation is addressed comparing the proposed vehicle trajectory against a baseline path. The error value represents the vehicle pose's drift.

The trajectory-related metrics are commonly implemented together with SLAM. As presented previously, SLAM can identify the vehicle pose to get the current trajectory (Prokhorov

et al. 2019). Other quantitative benchmark metrics used were success ratio (Bhusal et al. 2022) and execution time (Zhao et al. 2021b; Perez-Grau et al. 2021). The success ratio indicates how often the vehicle achieves the expected behavior. The execution time indicates how long it takes to solve the problem in simulated and real-world scenarios.

A learning curve was used for articles with learning-based methods (Zhao et al. 2021a; Grando et al. 2020; Peake et al. 2020; Delamer et al. 2021), illustrating the outcome's evolution over time. Other learning methods use deep learning metrics for problems such as classification (Zaki et al. 2017) and object detection (Albanese et al. 2022; Narazaki et al. 2022). As the methods implemented differ, the error rate was applied differently, i.e., the expected trajectory, heading angle, velocity, and acceleration, among others.

In contrast, other articles include a qualitative evaluation. They highlight the aspects where the proposal performed well and those where it did not. The qualitative assessment discussed aspects such as the performed trajectory, the visual recognition characteristics, and overall behavior stability. Lee and Kim (2022) enhanced the metric results by describing the flight route, vehicle, and target velocity in three different situations. A hyperparameter value was scaled in each situation, showing the control behavior for a refueling problem. Kumar et al. (2020) described the situation when the vehicle's camera recognizes the object of interest. In general terms, the qualitative comments allowed for an evaluation of each method's scope and limitations.

3.4.1 Experimental environment

Regardless of the type of metrics, each paper presented a custom experimental setup. This review divides the environments into Mathematical, Indoor, and Outdoor spaces. This is further subdivided into simulated and real-world scenarios. The mathematical space was interesting, with twelve papers that modeled the environment with Cartesian coordinates (Delamer et al. 2021; Zhao et al. 2021a). This kind of environment establishes area limits and the obstacle's position. Under this kind of experimental setup, how well the vehicle can avoid an obstacle and reach a given point can be observed. Other environments were in outdoor and indoor configurations, with 37 and 25 articles, respectively. In the indoor scenarios, more structured environments have corridors and doors, as in Prophet and Trommer (2020); Perez-Grau et al. (2021). Those scenarios presented a shaped trajectory similar to an L or an S. Conversely, in the outdoor scenarios, the trajectory depends on the altitude of flight and other landscape characteristics (Bhusal et al. 2022; Fu et al. 2022). For example, a forest area differs greatly from a city, or from other scenarios between those two. A summary of the environments can be found in Table 17.

Regarding the experimental setup, 28 papers used simulated and real-world scenarios. Most of them only described the main aspects, such as dimension, initial vehicle position, and target position. A brief description of the size and building characteristics were included in the real-world scenes. In terms of vehicle, the brand, the model, and a brief description of commercial drones (Bachrach et al. 2010; Dang et al. 2020; Fuentes et al. 2014; Konrad et al. 2017; Li et al. 2018; Nakamura et al. 2016, 2017; Perez-Grau et al. 2021; Tsiourva and Papachristos 2020; Yu et al. 2017) were included. All articles with real-world tests include the drone sensors' brand and model. Nevertheless, many drones were custom-built with specific parts for research purposes. Only seven articles (Al-Jarrah et al. 2013; Bachrach et al. 2010; Bhusal et al. 2022; Cho et al. 2022; Cocchioni et al. 2014; Fuentes et al. 2014; Mansouri et al. 2020b) mention the WiFi standard communication interface to transfer data between the vehicle and a ground station. Additionally, only Bachrach et al. (2010) mention the battery

Table 17 Reported environment used for autonomous navigation problem benchmarking

Environment	Type	Articles
Indoor: House, office, or any other building	Simulated	Bershatsky and Johnson (2013); de Plinval et al. (2017); Xu et al. (2013); Yu et al. (2017); Zaki et al. (2017); Zhao et al. (2021b)
	Real-world	Abdulov et al. (2022); Al-Jarrah et al. (2013); Bachrach et al. (2010); Capi et al. (2021); Chen and Lu (2022); Cocchioni et al. (2014); Dang et al. (2018, 2020); Fuentes et al. (2014); Alarcon et al. (2021); Li et al. (2018); Mansouri et al. (2020b, a); Nakamura et al. (2016); Nitsche et al. (2020); Perez-Grau et al. (2021); Prophet and Trommer (2020); Tsiourva and Papachristos (2020)
Outdoor: City, forest, and any open space	Simulated	Fu et al. (2013, 2022); Grando et al. (2020); He et al. (2017); Helgesen et al. (2019); Kusnur et al. (2021); Lugo-Cárdenas et al. (2017); Mao et al. (2020); Meng et al. (2019); Michaelsen et al. (2011, 2012); Nagarami et al. (2020); Narazaki et al. (2022); Peake et al. (2020); de Plinval et al. (2017); Sarmiento and Murphy (2018); Xu et al. (2013); Yang et al. (2021); You et al. (2020); Yu et al. (2017); Zaki et al. (2017); Zhao et al. (2021b); Zhu et al. (2018); Hu et al. (2022)
	Real-world	Bachrach et al. (2010); Bhusal et al. (2022); Cho et al. (2022); Cocchioni et al. (2014); Konrad et al. (2017); Li and Hu (2021); Louali et al. (2017); Mumuni et al. (2022); Nakamura et al. (2017); Prophet and Trommer (2020); Song et al. (2022); Sun et al. (2019); Wang et al. (2017)
Math: Mathematical function space	Simulated	Cao (2015); Delamer et al. (2021); Jia et al. (2022); Kumar et al. (2020); Lee and Kim (2022); Parfiryev et al. (2020); Tymochko et al. (2020); Yousuf et al. (2022); Zhang et al. (2017, 2018); Zhao et al. (2021a)

lifetime of the vehicle, with a 6000mAh lithium polymer battery for 15 minutes of flying autonomy.

Nevertheless, there is no common configuration in the environment organization or extension. In this regard, an application type of the environment was categorized in Table 18 to understand the method's objective better. The arbitrary environments are those presented for method testing purposes only, without specific details. These environments are commonly designed without obstacles or structures, and only present the target location. A structured environment is also arbitrary but with some defined characteristics like walls or corridors, limiting the navigation to certain areas. A known environment means a complete map is presented as prior information to the method. Pathway environments are those where the vehicle must follow a trajectory to reach target points. Other environments are landmark-based, from which some reference visual information is used to orient the vehicle. Dynamic and static environments were also defined as categories. A dynamic environment has different moving objects inside the scene, such as humans or other vehicles. Static environments are those where the only moving object is the flying vehicle. In this regard, only seven papers (Bhusal et al. 2022; Chen and Lu 2022; Cho et al. 2022; Lee and Kim 2022; Mumuni et al. 2022; Narazaki et al. 2022; Hu et al. 2022) report that the environment was dynamic, and all the other environments were considered static.

In general terms, no standard method or error metric application was found in all the papers. Each proposal presented its environment for a specific application. The wide variety of aspects makes it difficult to compare each approach fairly. The only exceptions where the proposals implemented in existing competition with real-world tests such as the International Aerial Robotics Competition 2009 (Bachrach et al. 2010) and 2011 (Bershadsky and Johnson 2013), the American Helicopter Society MAV Challenge (Nakamura et al. 2016), and the most recent AEROBOT-2021 UAV Challenge (Abdulov et al. 2022).

As with the experimental setup, there was no standard software used across all articles to run the simulated experiments, with each task presenting unique simulation requirements. Some papers used the RotorS simulation software, a Gazebo-based software focused on developing aerial vehicle research. The Gazebo software has been widely used in different robotic simulation projects. It is a simulator with broad community support, including for the development of simulated environments. However, most proposals used custom-developed or competition simulation software (Bershadsky and Johnson 2013; Nakamura et al. 2016) to execute their performance tests. Table 19 shows the main software categories found.

For object-goal navigation using UAV, the simulation software must correctly mimic real-world physics and vehicle dynamics. The simulation software must also provide a library to gather simulated sensor data and control the simulation. The use of competition problems makes it possible to standardize the simulation scenario. Competition define the information availability, system dynamics, and aspects evaluated. As such, a simulation software's main requirement is the capacity to mimic a real-world problem. The imitated details and simulation complexity must be able to be executed in a computing system.

Simulation software allows for the computation of a replica of the real-world. The auxiliary software included in the papers allows for communication between the proposed method and the simulated environment. One of the most common software applications is MATLAB, a programming and numeric computing platform with an integrated development environment. The programming language used depends highly on the simulation software employed. This language makes it possible to read the sensor data and operate through the simulated actuators. Another common software is the Robot Operating System (ROS), a communication middleware highly used in developing robotics solutions. The main advantage of ROS is the encapsulation of pieces of different programming language code under a single

Table 18 Environment application for each article found on the SLR

Type	Articles
Arbitrary: custom made for the problem	Abdulov et al. (2022); Al-Jarrah et al. (2013); Bachrach et al. (2010); Bershadsky and Johnson (2013); Bhusal et al. (2022); Cao (2015); Capi et al. (2021); Chen and Lu (2022); Cho et al. (2022); Cocchioni et al. (2014); Dang et al. (2018, 2020); Delamer et al. (2021); Furet al. (2022); Fuentes et al. (2014); Grando et al. (2020); He et al. (2017); Helgesen et al. (2019); Alarcon et al. (2021); Jia et al. (2022); Konrad et al. (2017); Kumar et al. (2020); Kusnur et al. (2021); Lee and Kim (2022); Li et al. (2018); Li and Hu (2021); Lough et al. (2017); Lugo-Cárdenas et al. (2017); Maitra et al. (2016); Mansouri et al. (2020b, a); Mao et al. (2020); Meng et al. (2019); Michaelsen et al. (2011, 2012); Mumuni et al. (2022); Nagrani et al. (2020); Nakamura et al. (2016, 2017); Narazaki et al. (2022); Nitsche et al. (2020); Parfiryev et al. (2020); Peake et al. (2020); Perez-Grau et al. (2021); de Plinval et al. (2017); Prophet and Trommer (2020); Song et al. (2022); Sun et al. (2019); Tsiourva and Papachristos (2020); Tymochko et al. (2020); Wang et al. (2017); Xu et al. (2013); Yang et al. (2021); You et al. (2020); Yousuf et al. (2022); Yu et al. (2017); Zaki et al. (2017); Zhang et al. (2017, 2018); Zhao et al. (2021a, b); Zhu et al. (2018); Hu et al. (2022)
Dynamic: other moving elements are present	Bhusal et al. (2022); Chen and Lu (2022); Cho et al. (2022); Lee and Kim (2022); Mumuni et al. (2022); Narazaki et al. (2022); Hu et al. (2022)
Known: prior information is presented	Cocchioni et al. (2014); Nitsche et al. (2020); Peake et al. (2020); Prophet and Trommer (2020); Tymochko et al. (2020); Yang et al. (2021); You et al. (2020); Zhang et al. (2018)
Landmarks: present visual cues for processing	Abdulov et al. (2022); Cho et al. (2022); Cocchioni et al. (2014); Fuentes et al. (2014); Jia et al. (2022); Mao et al. (2020); Meng et al. (2019); Michaelsen et al. (2011, 2012); Nakamura et al. (2016, 2017); Nitsche et al. (2020); Perez-Grau et al. (2021); Sun et al. (2019); Xu et al. (2013); Yu et al. (2017)
Pathway: trajectory path following	Grando et al. (2020); Parfiryev et al. (2020); de Plinval et al. (2017)
Structured: present elements, such as walls and corridors	Abdulov et al. (2022); Bershadsky and Johnson (2013); Dang et al. (2018, 2020); Fuentes et al. (2014); Alarcon et al. (2021); Mansouri et al. (2020b, a); Narazaki et al. (2022); Prophet and Trommer (2020); Song et al. (2022); Tsiourva and Papachristos (2020); Zhao et al. (2021b); Hu et al. (2022)

All applications were considered static unless stated to be dynamic applications

Table 19 Software used in autonomous navigation problem benchmarking

Software	Articles
Custom: custom or competition simulation software	Bershadsky and Johnson (2013); Bhusal et al. (2022); Cao (2015); Delamer et al. (2021); He et al. (2017); Louali et al. (2017); Mansouri et al. (2020b); Meng et al. (2019); Nakamura et al. (2016); Narazaki et al. (2022); Parfiryev et al. (2020); Peake et al. (2020); Perez-Grau et al. (2021); de Plinval et al. (2017); Prophet and Trommer (2020); You et al. (2020); Zhang et al. (2018); Zhu et al. (2018); Hu et al. (2022)
Gazebo-based: Gazebo and RotorS simulation software	Abdulov et al. (2022); Chen and Lu (2022); Cho et al. (2022); Dang et al. (2018); Fu et al. (2022); Grando et al. (2020); Alarcon et al. (2021); Nitsche et al. (2020); Yu et al. (2017); Zhao et al. (2021b)
Auxiliary: ROS and MATLAB	Capi et al. (2021); Chen and Lu (2022); Cocchioni et al. (2014); Dang et al. (2018); Grando et al. (2020); Alarcon et al. (2021); Kumar et al. (2020); Mansouri et al. (2020b); Nitsche et al. (2020); Perez-Grau et al. (2021); Song et al. (2022); Sun et al. (2019); Yousuf et al. (2022); Yu et al. (2017); Zhang et al. (2018); Zhao et al. (2021b); Zhu et al. (2018)
Specific: Google Earth, MAV3DSim, and other specific software	Lugo-Cárdenas et al. (2017); Mao et al. (2020); Michaelson et al. (2011, 2012)
Unknown: no software reported	Al-Jarrah et al. (2013); Bachrach et al. (2010); Dang et al. (2020); Fuentes et al. (2014); Helgesen et al. (2019); Jia et al. (2022); Konrad et al. (2017); Lee and Kim (2022); Li et al. (2018); Li and Hu (2021); Mansouri et al. (2020a); Mumuni et al. (2022); Nakamura et al. (2017); Tsiourva and Papachristos (2020); Wang et al. (2017); Xu et al. (2013); Zhang et al. (2017); Zhao et al. (2021a)

communication interface. This grants modularity to the system design, even allowing for parallel communication between simulated and real environments.

Simulated environments are sometimes sufficient to develop methods for solving further real-world problems. In most cases, they must be refined or aided with other methods in the real-world environment. Therefore, some papers used specific software to run their tests, such as Google Earth (Michaelsen et al. 2011, 2012) and MAV3DSim (Lugo-Cárdenas et al. 2017). Regarding software, Song et al. (2022) uses the Unreal Engine as the physics simulation engine, the same used by Fu et al. (2022) within the AirSim simulation platform. The mentioned software improves the visual representation of the simulated environment. Regardless of how much it may differ from the real environment, a simulation will improve the development of robotics solutions. Additionally, synthetic data has also been used to reduce the gap in implementing real-world solutions (Ju et al. 2022).

3.5 Differences over methods

This research question (RQ5) was mainly focused on finding differences in each paper's proposed methods. A main proposal comparison in Tables 20 and 21 show a comparison of the main proposals for the papers that solve the four tasks. This restriction was applied because these are essential tasks in UAV-based autonomous object-goal navigation. Only thirteen papers solved the control, planning, mapping, and perception tasks. The *UAV* column indicates the UAV type. All methods use a quadcopter. The *RW?* column indicated whether or not the paper presented a real-world implementation with "Y"es or "N"o. The *Env&Goal* column indicated the environment and goal types. An "(S)" or "(D)" indicated whether it was applied to a static or dynamic environment, respectively. The planning methods with an "(E)" represent an exploration-based approach, while the mapping methods with an "(L)" indicate a localization-based approach.

Most studies preferred to use classical feedback controllers such as PID, sophisticated model predictive controllers (MPC), and hardware-based Autopilot boards in their control methods. Classical feedback controllers are simple with an intuitive concept implemented in various applications that have low computational overhead. Nevertheless, they are unsuitable for complex systems with multiple inputs, outputs, and constraints. MPC can deal with complex systems by predicting future system behaviors and handling constraints, disturbances, and uncertainties. However, these controllers require an accurate system dynamics model and are more computationally demanding than PID. Finally, Autopilot boards such as PX4 can perform precise and accurate control under real-time constraints. Their implementation, however, is costly and complex, requiring specific hardware components and specialized knowledge.

For planning, some studies used the simplest hand-designed waypoint following, where a sequence of coordinates that the vehicle must reach have to be specified. Other studies used mainly graph- or tree-based methods, such as Dijkstra or Rapidly Exploring Random Tree (RRT). While it is true that both methods can be efficient and flexible in discrete or continuous domains, graph-based methods can be memory intensive in large state spaces, and tree-based methods can get stuck in local minima. Additionally, in cases where the target location is unknown, both types of methods do not guarantee a globally optimal path because they rely on heuristic guidance.

Most mapping methods create a surrounding representation by dividing the observed space and indicating the occupancy level of each cell or unit of representation. Occupancy grids and grid maps are the simplest types, and they can efficiently update and query occupancy

Table 20 Relevant methods overview, part 1

Article	Year	RW?	Env&Goal	Methods Control	Planning	Mapping	Perception
Abdulov et al. (2022)	2022	Y	(S)Indoor: object goal	Ardupilot	Dijkstra + A* algorithm	Semantic map + Visual SLAM (L)	CV + YoloV5
Narazaki et al. (2022)	2022	N	(D)Outdoor: no goal	Vehicle inner controller	Custom method	ICP-based mapping + SLAM (L)	Structure Motion (SfM) + CV
Zhao et al. (2021b)	2021	N	(S)Indoor; Outdoor: point goal	Vehicle inner controller	Intermediate goal	V-SLAM + Grid map + V-SLAM (L)	Spatial-structure information
Alarcon et al. (2021)	2021	Y	(S)Indoor: object goal	PX4	RRT* + Wavefront Raycast (E)	Voxblox + PX4 (L)	YoloV4
Delamer et al. (2021)	2021	N	(S)Math: point goal	Linear feed-back guidance law	Custom safe path planning	Occupancy map + Sensor modeling (L)	Kalman filter for state estimation
Peake et al. (2020)	2020	N	(S)Outdoor: object goal	Deep reinforcement learning	Advantage actor critic (E)	Quad-tree decomposition	ANN
Kumar et al. (2020)	2020	N	(S)Math: object goal	PD controller	Custom waypoint algorithm	Occupancy grid + SLAM (L)	CV

Table 21 Relevant methods overview, part 2

Article	Year	RW?	Env&Goal	Methods Control	Planning	Mapping	Perception
Dang et al. (2020)	2020	Y	(S)Indoor: No goal	Model predictive controller	Rapidly-exploring Random Graph (E)	3D map + multi-modal SLAM (L)	YOLO + bayesian filter
Tsiourva and Papachristos (2020)	2020	Y	(S)Indoor: No goal	Model predictive control	Graph-based exploration planner (E)	Vision-inertial (L)	CV + saliency map
Dang et al. (2018)	2018	Y	(S)Indoor: No goal	Receding horizon sampling	Semantical-enhanced RRT (E)	Occupancy map + Vision-inertial (L)	YOLO + Pretained classifier
Cocchioni et al. (2014)	2014	Y	(S)Indoor, Outdoor: Point goal	Autopilot	Waypoint following method	Optical flow + fast-SLAM (L)	CV
Bershadsky and Johnson (2013)	2013	N	(S)Indoor: Object goal	Vehicle inner controller	Dijkstra + Graph-based frontier (E)	SLAM (L)	CV
Bachrach et al. (2010)	2011	Y	(S)Indoor, Outdoor: Object goal	PID controller	Belief roadmap + Frontier-based (E)	Coordinate ascent search + SLAM (L)	Extended kalman filter + Two classifiers

information. Nevertheless, they may not be able to capture sufficient detailed information and are unsuitable for varying resolutions or irregular environments. On the contrary, Voxblox and Quad-tree decomposition can create multi-resolution mapping, with the latter being adaptive. However, both require significant computational resources and may suffer from discretization or in dynamic environments with varying occupancy. In general, all of these methods will require a higher amount of memory for large mapping spaces. Additionally, the Simultaneous Localization and Mapping (SLAM) technique can be implemented over any of those methods to locate the UAV on the spatial representation. Most of the time, SLAM is aided by Visual-inertial methods for tracking the UAV path within the map.

Most perception methods use computer vision (CV) techniques to extract useful features about the surroundings. Some use handcrafted extracted features by specifically designed kernels, similar to Spatial-structure information. These methods can deliver enough information about the processed sensor; however, developing such kernels requires considerable expertise. Others use pre-trained classifiers or object detection models such as YOLO to deliver insightful information for the mapping or planning stages. The main advantage of this is that useful extracted information can be learned from previously presented data; nevertheless, that data must be prepared and labeled to correctly extract useful features.

Another aspect was the problem formulation presented in Table 22. Most of the articles tackle UAV control as a navigation-only problem, where the vehicle must move around the environment safely. However, this does not mean that the proposed methods solve the problem autonomously. Aerial-only formulation problems, such as landing and aerial refueling, were also found. In aerial-only problems, the proposed methods use visual sensors to identify the landing or fueling location they must reach to accomplish the mission. Formulating the problem as an object search was the most suitable method for object-goal navigation. Not all methods deal with the problem autonomously, although they can navigate an area searching for objects of interest. A few papers formulated the problem as exploration, attempting to move around in unknown areas. They aim to gather information on the surroundings and to better choose where the target might be. Another problem formulation includes no-control methods, mainly to recognize different elements or objects within the scene, such as landmarks or targets. Finally, other variant problem formulations were identified, such as algorithm optimization (Zhang et al. 2018), city mapping (Yang et al. 2021), position stabilization (de Plinval et al. 2017), runway detection (Nagarani et al. 2020), state estimation (Konrad et al. 2017), battery lifetime extension (Cocchioni et al. 2014), and remote guidance (Louali et al. 2017).

Another difference found between each proposal was the definition of the objective. While most proposals formulate the problem as a navigation problem, three main categories of objective definition were found. The first was point goal, which defines the target as a landmark at specific coordinates that the vehicle must reach. The second was object goal, where an object is defined as a target, and the vehicle's final destination is the object's location. The articles which fit into each category are shown in Table 23. However, most of the articles did not specify the type of goal employed, while some did not include vehicle control or the authors did not report it.

4 Challenges and future strategies

The research questions from this SLR allow for a better understanding of the problem. However, a future steps analysis is essential to understand what is still missing. Future studies

Table 22 Articles's problem formulation discovered on the SLR

Problem	Articles
Navigation: area navigating problem	Abdulov et al. (2022); Bachrach et al. (2010); Cao (2015); Capi et al. (2021); Fu et al. (2022); Grando et al. (2020); Kumar et al. (2020); Mansouri et al. (2020b,a); Mao et al. (2020); Michaelson et al. (2011); Mumuni et al. (2022); Narazaki et al. (2022); Parfiryev et al. (2020); Song et al. (2022); Zaki et al. (2017); Zhao et al. (2021b); Hu et al. (2022)
Aerial only: landing and aerial refueling problem	Cho et al. (2022); Lee and Kim (2022); Li and Hu (2021); Maitra et al. (2016); Meng et al. (2019); Nakamura et al. (2016, 2017); Sun et al. (2019); Xu et al. (2013); Yu et al. (2017); Zhu et al. (2018)
Search: object searching problem	Bershadsky and Johnson (2013); Dang et al. (2020); Alarcon et al. (2021); Peake et al. (2020); Wang et al. (2017); Yousuf et al. (2022)
Control: vehicle control problem	Al-Jarrah et al. (2013); Bhusal et al. (2022); Perez-Grau et al. (2021); Prophet and Trommer (2020); Zhao et al. (2021a); Hu et al. (2022)
Path planning: trajectory planning problem	Chen and Lu (2022); Delamer et al. (2021); Jia et al. (2022); Kusnur et al. (2021); Lee and Kim (2022); Lugo-Cárdenas et al. (2017); Mao et al. (2020); Zhao et al. (2021b)
Tracking: target tracking problem	Al-Jarrah et al. (2013); Fu et al. (2013); He et al. (2017); You et al. (2020); Zhang et al. (2017); Hu et al. (2022)
Exploration: area exploring problem	Dang et al. (2018); Alarcon et al. (2021); Tsiourva and Papachristos (2020)

Table 23 Goal type used in autonomous navigation problem definition

Goal type	Articles
Object goal: locate a desired object	Abdulov et al. (2022); Al-Jarrah et al. (2013); Bachrach et al. (2010); Bershadsky and Johnson (2013); Fuentes et al. (2014); Alarcon et al. (2021); Kumar et al. (2020); Nakamura et al. (2016); Peake et al. (2020); Yousuf et al. (2022)
Point goal: reach specific coordinates or position	Capi et al. (2021); Chen and Lu (2022); Cho et al. (2022); Cocchioni et al. (2014); Delamer et al. (2021); Fu et al. (2022); Grando et al. (2020); He et al. (2017); Jia et al. (2022); Kusnur et al. (2021); Lee and Kim (2022); Louali et al. (2017); Lugo-Cardenas et al. (2017); Mao et al. (2020); Nagamani et al. (2020); Nakamura et al. (2017); Parfiryev et al. (2020); Perez-Grau et al. (2021); Prophet and Trommer (2020); Wang et al. (2017); Xu et al. (2013); Yu et al. (2017); Zhang et al. (2017, 2018); Zhao et al. (2021a, b); Hu et al. (2022)
Unspecified: No target specified	Albanese et al. (2022); Bhusal et al. (2022); Cao (2015); Dang et al. (2018, 2020); Fu et al. (2013); Helgesen et al. (2019); Konrad et al. (2017); Li et al. (2018); Li and Hu (2021); Maitra et al. (2016); Mansouri et al. (2020b, a); Meng et al. (2019); Michaelsen et al. (2011, 2012); Mumuni et al. (2022); Narazaki et al. (2022); Nitsche et al. (2020); de Plinval et al. (2017); Sarmiento and Murphy (2018); Song et al. (2022); Sun et al. (2019); Tsiourva and Papachristos (2020); Tymochko et al. (2020); Yang et al. (2021); You et al. (2020); Zaki et al. (2017); Zhang and Duan (2014); Zhu et al. (2018)

reported in the literature identify two main scopes. One scope refers to how to improve the presented method, and another is about how to extend it. Because other studies did not mention future steps, an extension inference was made.

Only ten papers explicitly mention enhancement and extension of the proposal presented. The most recent ones wanted to address computational complexities to be more efficient (Mumuni et al. 2022; Narazaki et al. 2022). Initial solutions improvements and data-efficient methods are also considered (Delamer et al. 2021; You et al. 2020). Other authors reported that improvements in sensor data are required (Bhusal et al. 2022). Processing better state estimation considers robust sensor data for new real-world flight tests (Prophet and Trommer 2020). Additionally, robust visual-based methods are also required in aggressive flight behavior (Bachrach et al. 2010).

Eight papers point out future method enhancements, almost all sensor-related. From the most recent, Lee and Kim (2022) reported velocity information acquisition for temporal unavailable data. Additionally, Cho et al. (2022) reported enhancing no GPS-based methods for velocity and position information acquisition. Chen and Lu (2022) reported future improvements in visual-based state estimation during aggressive flight with data fusion. For Fu et al. (2022), refining the UAV kinematics characteristics should improve flight smoothness. Other authors referred to address limitations in sensors (Al-Jarrah et al. 2013; Bershadsky and Johnson 2013) and computational requirements (Yu et al. 2017; Xu et al. 2013).

In 26 articles, the authors discussed how to extend their proposals. Eight of these papers would like to use more complex scenarios (Hu et al. 2022; Capi et al. 2021; Mansouri et al. 2020b; Wang et al. 2017). Tests in 3D spaces (Yousuf et al. 2022; Li et al. 2018), unknown scenarios (Perez-Grau et al. 2021), and with more elements on it Peake et al. (2020) are also included. The other eight papers want to add real-world test cases (Michaelsen et al. 2012; Sun et al. 2019; Song et al. 2022), including a real fixed-wing UAV (Lugo-Cárdenas et al. 2017). Other cases want to increase the complexity of already real-world applications (Alarcon et al. 2021). Others want to improve generalization in vehicle (Grando et al. 2020) kind or include dynamic elements (Louali et al. 2017).

From the remaining papers, some authors also suggested sensor processing extensions. Two of them look to improve the pose estimation by certain adjustments. Li and Hu (2021) indicate using visual key points for position to pose autoland. Nitsche et al. (2020) indicate a pose uncertainty reduction for improving map-adjustment accuracy. Alternatively, Kusnur et al. (2021) and Mao et al. (2020) recommend dealing with sensor history and multi-sensor sources. Tymochko et al. (2020) and Tsiourva and Papachristos (2020) presented an interesting recommendation. The first is about an alternative to prior information in vision-based methods. In contrast, the second is for a better stationary surrounding objects prior information. Finally, Cocchioni et al. (2014) instructs the use of more positioning beacon sensors for real-time performance.

Final extensions suggest checking out more efficient models for the required tasks. For example, Albanese et al. (2022) mentions DroNet and EdgeNet for navigation and surrounding perception. Jia et al. (2022) points out the multi-agent approach in dynamic environments and real-time path planning. While Fuentes et al. (2014) and Fu et al. (2013) mention extending with object detection methods and other tracker algorithms, respectively. Mansouri et al. (2020b) address payload, flight time, and communication distance in real-world scenes.

Extra challenges were extrapolated from the SLR findings. It is noteworthy that only a few papers address the object-goal navigation problem. All papers were included in at least one category for macro-comparison. This SLR recommends addressing the four main tasks of control, planning, mapping, and perception under a methodology that allows reusing the

sensors processed information by the methods of each task. The unified methodology should improve the computational cost and the generalization of autonomous UAV applications. A secondary recommendation addresses developing a framework for UAV navigation solutions, which should include standard comparison metrics and an environment. Additionally, it must ensure experimental replication while considering safety, ethical, and legal aspects. An extra challenge is to measure the autonomy level between different proposals.

4.1 Autonomous control

The autonomous UAV concept implies a power of self-governance in the performance of the control function. In this regard, there is a difference between automatic and autonomous flight control, with the latter presenting a higher degree of reasoning. Automatic flight provides the necessary control to operate in normal scenarios. However, it requires manual adjustment in an unstructured environment. Conversely, an autonomous flight must detect and respond to unexpected events and conditions during control. Bio-inspired methods have been proven to achieve complex behavior. Nevertheless, this does not ensure full autonomy (Millán-Arias et al. 2022).

According to Chen et al. (2009), an autonomous control architecture presents three hierarchy levels with the principle of Increasing Precision with Decreasing Intelligence (IPDI) (Saridis 1989). The lowest is the execution level, with the fewest intelligence control algorithms. In the middle is the coordination level, which acts as an interface between both hierarchies, processing sensor data. At the top, is the organization level with the highest intelligence and lower precision, including all of the decision-making methods.

Autonomous control for object-goal navigation expands these hierarchy levels into five subproblems that must be solved, as presented in Fig. 5. Each subproblem involves more than one task. For example, the SLAM method combines localization and mapping to recognize objects and the vehicle location within the environment. The subproblems must be addressed in a cascade fashion, starting from the simplest, which must be solved in real-time, and finishing with the most complex, which requires semantic information processing.

4.1.1 Spatial interpretation

The spatial interpretation problem, as an extension of localization, was defined as recognizing the vehicle's current position using internal information only. As mentioned earlier, perception is the fundamental task that processes all sensory data to provide insightful information to the other tasks. Hence, the perception task must process specific types of sensors to deliver

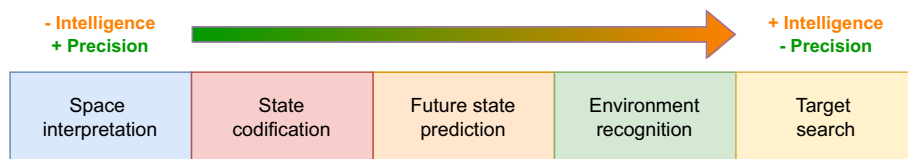


Fig. 5 Challenges to be solved in order to achieve autonomous behavior in object-goal navigation using UAV. The simplest problems are on the left and the more complex ones on the right. Each subsequent problem requires the previous one to be solved

correct information so that the localization task can achieve an accurate position estimation. Estimating the current position facilitates the computation of how the state control must change to reach the next desired position.

The localization concept is then expanded into spatial interpretation, because the vehicle position can be interpreted as either absolute coordinates or as a relative distance. Therefore, spatial interpretation must allow the vehicle's movement to be observed through the sensor data measurements and by recognizing the difference between one location and another within an environment. From the SLR, it can be inferred that the IMU and magnetometer are suitable sensors to solve this problem. However, further research must be carried out to confirm this. Additionally, metrics are required to measure how well this problem has been solved, and ground-truth data may need to be obtained to compare against experimental results.

4.1.2 State codification

Because the control task requires state variables, state codification becomes another relevant problem for achieving autonomous behavior. Again, the perception task plays an essential role in processing the sensor data and attempting to negotiate a unified codification of state variables with the control task. This codification must be suitable for use by the control task, which will trigger the various vehicle actuators or motors. The state codification should contain all the information required by the control task to achieve movement. The control task requires information on the current and the target states in order to achieve any movement. If no goal state is given, the vehicle must keep still.

Sensors and metrics must need to defined to measure how well this problem is being solved and how efficient the solution is. The state codification discovered in this SLR concluded that the UAV's sensor data of movement and position appear to be suitable to solve this problem. In terms of metrics, the evaluation of the state codification must be compared to baseline trajectory. However, further experiments are required.

4.1.3 Future state prediction

For the vehicle to perform any movement, it requires a target state given by the planning task, ante further processed by the control task. This future state estimation must be addressed to allow the vehicle to move around, maximizing its efficiency in traveling from one point to another. With regard to the navigation behavior, the travel plan must reach a given location fast and safely. For exploration behavior, the travel trajectory should select unknown locations to explore the environment. A state codification system interprets where the vehicle is in the environment. With this information, the planning task must ensure that the control task reaches all states defined in the routing schema until a final state is reached.

The sensor data required to solve this problem of state codification and spatial interpretation should capture the UAV's current location and infer future states from it. An assessment metric for this problem must analyze the similarities between the proposed trajectories, and how far the vehicle traveled to reach a given location.

4.1.4 Environment recognition

Another important problem that must be solved is the recognition fo all possible elements within the environment. This problem should be addressed by the perception and mapping

tasks, which process the sensor information to identify the objects and location. First, the perception task must recognize the object features and their labels, which can then be further processed by the mapping task. This mapping process uses the perception task's information to assign the object data to a reference method, combining it with the current vehicle location and heading angle.

Object recognition in navigation requires visual data to identify an object's visual characteristics, assigning a label to what is being observed. Depth information is also required to correctly identify how far the object is from the vehicle's current position. Metrics that can measure the quality of the solution to this problem can be object detection metrics, such as the mean average precision of distance error measurements.

4.1.5 Target search

The most important problem to be solved by a vehicle under object-goal navigation is the location of the target. For this, the mapping and planning tasks must work towards processing the target location. The mapping task must deliver useful information about the vehicle's surroundings to the planning task. The planning task will then trace a route to an area where the target is likely to be found. Selecting an area can be challenging because a wrong direction or semantic understanding will cause the vehicle to move away from the objective's location. In both indoor and outdoor environments, it will drain the UAV's battery, limiting the available search time. Another problem relates to how the vehicle will recognize which object must be found.

The sensory data required to solve this problem are those related to the environment recognition and spatial interpretation problems. A metric for quality measurement of this problem must be investigated, one possibility being the final sum of all of the aforementioned problems.

4.2 Comparison metrics

As mentioned earlier, the lack of a standard metric to fairly compare each method presents a challenge in establishing the state-of-the-art. A previous navigation problem evaluation article presented by Anderson et al. (2018) acknowledges context evaluation. The context must specify a goal and sensory input, exploration exposure, and some recommended metrics. These recommendations were proposed for ground embodied agents. However, they seem suitable to be implemented in UAV agents. The metrics recommended are:

- A form of indicating that the agent reached the goal position.
- Using a geodesic distance to measure the difference between objects within the space.

In this regard, this document followed the two-goal specifications referred to as point goal and object goal. Definition of the goal makes it possible to compute the outcome once the agent indicates that the objective has been reached. In the case of a point or an object goal, the successful distance presents a threshold value, and the authors' recommendation is 2x the size of the agent's body. In Anderson's paper, other interesting navigation problem solution for embodied agent recommendations (Anderson et al. 2018) can be found.

Although no single standard metric was found in the articles of this SLR, the benchmark values can be used as the initial metric. As mentioned above, a single value fails to represent a paper's autonomy level. Therefore, a 5-component autonomy score (*AS*) is proposed based

on the problems mentioned in the previous section:

$$AS = SI + SC + FS + ER + TS, \quad (1)$$

with SI being spaatial interpretation, SC being state codification, FS being future state prediction, ER being environmental recognition, and TS being target search scores.

The SI term defines the error estimation in the sensor data. A suitable error function should measure the quality of approximation to sensor information. Raw sensor data can be noisy and measured under a specific time interval. Approximation to a real sensor value reduces presence of outliers and ensures data acquisition. For example, Chen and Lu (2022) applied the Kalman filter to noisy density-based spatial clustering obtained from a LiDAR sensor. The KF delivers an approximated LiDAR measurement from historical sensor values. The difference between the approximation and ground truth values determines the method's quality. Absolute error metrics should be useful in these cases.

The SC term defines how well a method translates the sensor data into a control state. A vehicle's control state depends on its motion dynamics or kinematic definition. Different approaches have used the vehicle's position or pose within the global frame, or analogous use-related change rates such as velocity and attitude. The state estimation is derived from local frame sensor data processing. Some methods obtain estimations from different sensor data sources. For example, Song et al. (2022) estimated position and velocity from LiDAR and monocular RGB images. A comparison of estimates against ground-truth data makes it possible to compute the RMS or RPE values.

The FS term indicates the quality of the chosen target states that the vehicle must reach. Path planning methods instruct the UAV to reach a given future state. In this regard, comparisons between the vehicle's chosen trajectory and ground truth are suitable. This metric can consider the absolute relative distance, presented by Narazaki et al. (2022). It can also consider the number of points, as presented by Tymochko et al. (2020), or the absolute trajectory error, proposed by Mumuni et al. (2022). Besides the distance, the FS term should consider some exploration-related aspects. For example, the coverage area rate presented by Prophet and Trommer (2020) or the target distance rate presented by Zhang et al. (2018).

The ER and TS are high complexity terms that are similar to deep learning metrics. The ER term indicates how good the surrounding representation is. The environmental representation comprises all of the obstacles, free spaces, and objects of interest. Its complexity derives from the semantic information acquisition for a global map. Dang et al. (2018) propose a modified version of the occupancy grid map, including the object present. A pre-trained classifier allowed the insertion of semantical information on the cubical voxel. Nevertheless, as mentioned by Abdulov et al. (2022), semantic maps are still challenging. Since it is a data association-related problem, tracking metrics presented by Chen and Lu (2022) for multiple objects seems suitable.

TS , similar to ER , requires a semantic understanding of the environment. The main difference lies in the purpose of the semantic information. TS evaluates how well the decision-making process was able to find the target object. Perez-Grau et al. (2021) used a success ratio and mission time as comparison metrics. Decision-making assessment can be done through a weighted version that contains both elements. The success ratio indicates how often the mission was accomplished. The time indicates how long it took to complete the mission.

4.2.1 Explainability

Beyond metric selection, explainability is an additional tool that can help UAV behavior be more easily understood. Explainability refers to how a method processes information

to give an outcome in a form humans can understand (Ayala et al. 2021). Explanations of autonomous systems strengthen human-AI interaction effectively and efficiently (Holzinger et al. 2020). Parallelling the same five previously-mentioned components, five explanations must be given by the vehicle, as follows:

1. **Where is it currently located?** Seeks to obtain the vehicle's understanding of its current position.
2. **How is current control state codified?** Seeks to understand the variables used to control the vehicle.
3. **Where should it be next?** Seeks to interpret how the vehicle chooses one position over others.
4. **What is around it?** Seeks to visualize which aspects are relevant for the vehicle to recognize an interesting object.
5. **Where is the target object?** Seeks to interpret the decision-making process that chooses one region over another.

Different levels or additional explanations must be included for each scenario application, such as reactive, disposition, social, and cultural (Dazeley et al. 2021). These explanations should enhance the development of the methods, making them more efficient and allowing unexpected behaviors to be more easily debugged.

4.3 UAV Sim2Real

The development of robotic-based solutions is costly in terms of hardware, both for acquisition and for maintenance. Testing also presents a high risk of accident if any malfunction or unexpected behavior occurs. Therefore, modeling and testing in simulated scenarios is essential to prevent damage to equipment or personnel (Ayala et al. 2020). The simulation software selection becomes a challenge for the following reasons: Many applications address problems that are too specific or use custom software not publicly available. Additionally, experiments reported in studies using simulations lack details about the simulation setup. The main idea behind using a simulated testbed is the ability to replicate previous studies and thereby, achieving a fair comparison against other methods. However, this is not possible without information on the setup. Moreover, selection of a suitable simulation software must ensure its timely maintenance to prevent obsolescence, such as, the AirSim³ simulation software that will be archived.

Another big concern with regard to simulation software selection is its capacity to replicate the real-world. The better the replica is, the less it will cost to transfer the solution from a simulated to a real-world scenario. In UAV development, wind is a crucial factor in imitating the aerial dynamics of real scenarios. Other aspects that must be included are communication protocols, battery life, and sensor battery requirements. The communication protocol must be exhaustive, including onboard sensor data transfer, remote control operations, and contact between vehicles. Battery availability and drain rate must also be considered as constraints to the UAV's flight time. As a considerable part of the articles in this SLR used RotorS, a Gazebo-based simulation software, it can be considered an initial option. However, further research must be done.

Other software and development hardware must be considered part of a UAV development framework to reduce the gap between simulation and reality. A special consideration must be made for the graphics rendering software, particularly for the visual sensor information. A

³ <https://github.com/microsoft/AirSim>

more realistic appearance will make achieve visual-domain adaptation easier (Truong et al. 2021). Autopilot boards seem to be a suitable option to use in both simulated and real-world scenarios. Besides having a wide development community, they can be used as software-in-the-loop or hardware-in-the-loop, making their integration into any vehicle control project easy.

The ROS software has great features in the robotic development area, including useful pieces of ready-to-use code. ROS allows for communication between each robot's sensor and actuator, even if they were developed in different programming languages. Another advantage of ROS is that some robot and AV manufacturers adopt it for their products, making the use and testing of commercial vehicles even easier.

4.3.1 Safety considerations

Since many UAV applications involve environments that contain people, safety aspects must be considered in the event of a system failure. Almost none of the surveyed articles included this consideration in their method proposal. In this regard, Afman et al. (2018) propose a safe-to-crash UAV design in both control logic and mechanical constraints. This UAV safety paradigm was used with an automatic control method as well as with remote control. The logic control includes operations related to data error tolerance and trusted communication. The mechanical constraints address physical contact with unexpected forces, caused by a system fault. Balestrieri et al. (2021) addressed the safety topic under different maneuver phases, classified into seven categories detailed in the referenced paper. The authors established a relationship between the category and a leading sensor required to mitigate the safety issue effectively. Although there is information in the literature about safety in UAV development, it remains to be effectively applied.

4.3.2 Legal and ethical implications

As UAV applications increase each day, they show a positive impact on society, being helpful for smart cities' purposes (Mohamed et al. 2020). With their increasing use in civilian scenarios, ethical and legal issues need to be emphasized. Since a UAV can be easily deployed to capture images and data, it poses serious concerns regarding safety, privacy, and credibility, among others. Different papers already discussed the ethical implications of drone technology in journalism (Culver 2014) and surveillance activities (West and Bowman 2016).

Another interesting aspect is the legal concern related to the autonomous operation of UAVs. The website Drone Laws⁴ shows drone use legislation in different territories. Most require registering the drone and its respective specifications, in order to allow it to be used, with some space restrictions. However, all regulations are refer to a human UAV operator, and it remains to be evaluated how autonomous operation might be included. Legal and ethical aspects are still a challenge for the proposed methods and UAV solution development.

5 Conclusion

Over the past decade, an increasing number of UAV-based solutions have emerged in diverse areas of implementation and varied methodologies for achieving suitable applications. The

⁴ <https://drone-laws.com>

focus of this SLR was to identify actual development using UAVs under an object-goal navigation problem. Autonomous navigation can only be achieved if the UAV solves the tasks of control, planning, mapping, and perception without human intervention. Nevertheless, only some of the papers included address the navigation problem in an autonomous manner. Most papers solved only one task involved in autonomous navigation. The number of articles that address object-goal navigation was even fewer. Only a few papers included reaching the location of an object of interest. Using the knowledge extracted from the articles, it was possible to identify two main behaviors: exploration and navigation. The four tasks necessary to achieve an autonomous approach were also identified. The behavior depends on the processing form of each task. It was also possible to classify the methods used and proposed by each author. This classification considers the sensory data, state control definition, benchmark metrics, and problem formulation differences. Because of the lack of a standard test suite and development framework, no direct comparison between methods was performed. Moreover, real-world methods only report a few of the real constraints present in UAV control. Finally, a novel challenge was discovered during this review: the definition of standardized metrics, experimental setups, and development frameworks for the object-goal navigation problem using UAVs. With such standardization, future studies in the area will be easily reproducible and comparable.

Appendix A Methodology details

A.1 Review protocol

Once the research questions were defined, the review protocol used them to establish the search process. The search process started by defining a query string, after which, inclusion and exclusion criteria for papers were set. Quality assessment, data extraction, and analysis parameters were also outlined. Different pilot searches were executed during the protocol planning. This enabled an understanding of the results obtained through analysis of the most commonly used terms. These pilot searches were used to improve the research question, the query string, and the inclusion and exclusion criteria.

The search process was conducted in five digital libraries: ACM Digital Library,⁵ IEEE Digital Library,⁶ ISI Web of Science,⁷ Science@Direct,⁸ and Scopus.⁹ The terms used in the query string were derived from the research questions. They were further refined in five trial searches to find the ideal combination of terms to obtain the most relevant papers. The final query string executed on each library search engine was as follows:

("object goal" OR "object conditioned" OR "object directed" OR "object driven" OR "object oriented" OR "object search" OR "goal conditioned" OR "goal directed" OR "goal driven" OR "goal oriented" OR "goal search" OR "target conditioned" OR "target directed" OR "target driven" OR "target oriented" OR "target search") AND ("navigation" OR "exploration") AND ("UAV" OR "UAVs" OR "aerial" OR "agent")

An initial filter was applied to each database search, focusing on papers between 2011 and 2022. The primary papers were obtained using the query string on keywords, titles, and

⁵ <http://portal.acm.org>

⁶ <http://ieeexplore.ieee.org>

⁷ <http://www.isiknowledge.com>

⁸ <http://www.sciencedirect.com>

⁹ <http://www.scopus.com>

abstracts. After that, they were filtered using the following inclusion and exclusion criteria:

Inclusion Criteria (IC):

- Papers with control methods.
- Papers with a navigation task.
- Papers with an object navigation problem.
- Papers with an object recognition task.

Exclusion Criteria (EC):

- Duplicated papers.
- Papers in a language other than English.
- Papers in progress, posters, workshops, short papers or book chapters.
- Papers in the Neuroscience area.
- Papers with human-centered methods.
- Papers with multi-agent approaches.
- Papers with no UAV applications.
- Papers with no navigation task.
- Papers with no object search.

Each selected paper was given a score based on ten questions. The higher the score, the more valuable the paper was considered to be for this review. The complete question list is presented in Table 25. It addresses the article's primary goal, experimental setup, benchmarking, and reproducibility. Each question was answered with "yes" if all of the indicated aspects were met, scoring a value of 1. The answer was marked "partially" if some aspects were met, scoring a value of 0.5. Finally, the answer was "no" if none of the indicated aspects were met, scoring a value of 0. Once all questions were answered, the total value of each answer was determined to obtain its score. The quality question answer details are shown in Appendix A.3.

The review protocol also considered helpful data extraction details. This included publication information, methods, software, test scenarios, and metrics used for comparison. The entire extraction process was done manually, as well as the quality assessment of selected articles.

A complete list of extracted data and descriptions is shown in Appendix A.4. However, in an attempt to reduce potential bias, the review protocol was elaborated, reviewed, and refined by four out of five authors. The other author did not participate in the review elaboration and checked the document structure and the coherence between the protocol created and the obtained results.

A.2 Review execution

The execution of the query string on five different databases returned a total of 646 articles. ACM produced 9 of them, 104 were from IEEE Xplore, 114 from WebOfScience, 150 from Science Direct, and 269 from Scopus. The Parsifal online tool¹⁰ was used to execute the literature review. This simplified the articles' organization, detecting duplicates and extracting metadata. Parsifal is an online tool that facilitates the planning and carrying out of systematic literature reviews in the software engineering context. The tool's framework starts with protocol definition. This is where the objective and research questions must be

¹⁰ <https://parsif.al>

Table 24 Number of articles selected per stage

Database	Search stage	Filtering stage	Excl/Incl stage
ACM	9	9	0
IEEE Xplore	104	71	13
WebOfScience	114	12	1
Science direct	150	149	41
Scopus	269	223	12
Total	646	464	67

defined. Subsequently, the information for quality assessment and further data extraction are detailed. This allows the tool to create a form for each article containing one field per piece of data required. Once all of the forms are filled up, the data can be exported for processing and analysis. Finally, all of the articles were passed through Mendeley¹¹ for bibliographic organization. This digital library search was performed in November 2022.

The selection process was divided into three stages:

- The first stage was to obtain the list of articles from the digital library search by applying the query string mentioned earlier. In this stage, 646 articles published between 2011 and 2022 were obtained.
- The second stage was to filtering out any duplicated articles found. A total of 182 duplicate articles were removed, leaving 464 articles.
- The third and final stage was the application of the exclusion and inclusion criteria after reading each article's title and abstract. The exclusion criteria were applied first and papers outside of the review's scope were removed. The inclusion criteria were then applied to select the most relevant articles. Following application of the inclusion criteria, 67 articles were obtained for a more detailed analysis. Table 24 a detailed count of articles for each stage and database.

The quality assessment was performed on these 67 articles, based on the checklist and the answer score. A higher score meant that the article was more relevant to the review. The score was used to understand the results better, synthesize the data, and analyze the proposal. Table 25 shows the score summary. It includes the answers for each quality question (QQ) from the checklist. It can be seen that most of the answers were negative, with only three of them partial. This lack of a positive majority meant that the articles did not present sufficient information on the object-goal navigation problem, its methods, or its evaluation. Questions QQ1 and QQ2 identified whether the articles addressed the object-goal navigation problem and the experimental setup used. A negative answer and a partial answer were found in 59% and 75% of the papers, respectively. From these results, it was concluded that most articles do not include the object-goal navigation problem with sufficiently structured experimental cases. The following questions, QQ3 and QQ4, identified the implementation of validation scenarios, and categorized them as either simulated or real-world, respectively. QQ3 had 44.78% negative answers, demonstrating that at least half of the methods were tested in a simulated environment. Similarly, QQ4 demonstrated that only a small minority of the articles executed real-world tests. QQ5 and QQ6 identified whether the articles presented quantitative metrics and if they were compared against previous proposals. They returned 82.09% partial answers and 73.13% negative answers, respectively. QQ7 and QQ8 are related

¹¹ <https://www.mendeley.com/>

Table 25 Quality criteria

ID	Question	Y	P	N
QQ01	Is the object-goal navigation methodology described?	7 (10.45%)	20 (29.85%)	40 (59.70%)
QQ02	Is the scenario well documented?	14 (20.90%)	50 (74.63%)	3 (4.48%)
QQ03	Does it present a 3D simulated scenario?	16 (23.88%)	21 (31.34%)	30 (44.78%)
QQ04	Does it present a real world implementation?	19 (28.36%)	13 (19.40%)	35 (52.24%)
QQ05	Does it present benchmark metrics?	5 (7.46%)	55 (82.09%)	7 (10.45%)
QQ06	Does it present a comparison with other approaches?	8 (11.94%)	10 (14.93%)	49 (73.13%)
QQ07	Does it expose hyper-parameters required by the method?	16 (23.88%)	43 (64.18%)	8 (11.94%)
QQ08	Does it include learning algorithms?	14 (20.90%)	17 (25.37%)	36 (53.73%)
QQ09	Does it include a pseudo-code in the proposal explanation?	18 (26.87%)	5 (7.46%)	44 (65.67%)
QQ10	Does it have a code repository available?	3 (4.48%)	1 (1.49%)	63 (94.03%)

The values in bold represent the major number of answers per question

to algorithm implementation, while QQ7 expresses whether the required method parameters were detailed with 64.18% partial answers. QQ8 indicated whether the method presented was an end-to-end machine learning solution, with 53.73% negative results. Finally, QQ9 and QQ10 assessed the reproducibility level of each method. They returned negative results of 65.67% and 94.03% for pseudo-code guidance and publicly available code, respectively. More details regarding each answer level for each question and individual results can be found in Appendix A.3.

From the data extracted, the number of publications per country was obtained, as shown in Fig. 6.

A.3 Quality questions

The following questions were applied to obtain a score for each article's interest level. Each question presents three answer levels, according to how well the question is satisfied. The question list and answer level details were as follows:

QQ01: Is the object-goal navigation methodology described?

- ☐ Yes: The object-goal navigation problem's entire method and aspects were described.
- ☐ Partially: The methodology only describes a navigation problem.
- ☐ No: Neither object-goal navigation nor navigation methodology is described.

QQ02: Is the scenario well documented?

- ☐ Yes: All the scenario descriptions, including elements such as type, dimensions, objects present, were reported.
- ☐ Partially: Only a few aspects were reported.
- ☐ No: No environment details were reported.

QQ03: Does it present a 3D simulated scenario?

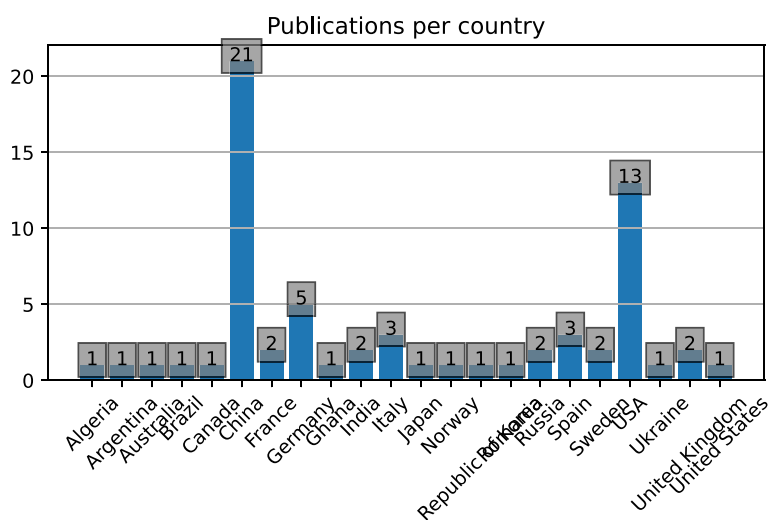


Fig. 6 Number of publications per country

- ☐ Yes: A fully 3D environment was used, with a physics engine.
- ☐ Partially: A 2D environment was used, with or without a physics engine.
- ☐ No: No simulated environment was reported.

QQ04: Does it present a real world implementation?

- ☐ Yes: The entire proposal was fully tested in a real-world environment.
- ☐ Partially: A dataset from a real-world scenario was used for proposal testing.
- ☐ No: No real-world data was used for testing.

QQ05: Does it present benchmark metrics?

- ☐ Yes: It uses quantitative and qualitative metrics.
- ☐ Partially: It uses only quantitative or qualitative metrics.
- ☐ No: No quantitative metrics were reported.

QQ06: Is the proposal compared with other approaches?

- ☐ Yes: The proposal is compared with at least one previous study.
- ☐ Partially: The proposal is compared against a defined baseline.
- ☐ No: The proposal does not report any comparison.

QQ07: Does it detail all of the hyper-parameters used?

- ☐ Yes: The results describe all of the method's hyper-parameters.
- ☐ Partially: The results only highlight a few hyper-parameters.
- ☐ No: No hyper-parameters were reported.

QQ08: Does it include learning algorithms?

- ☐ Yes: The method uses a learning algorithm as an end-to-end solution.
- ☐ Partially: The method uses a learning algorithm as part of the solution.
- ☐ No: The method does not use a learning algorithms.

QQ09: Does it include pseudo-code in the proposal explanation?

- ☐ Yes: The paper includes pseudo-code in the methodology.
- ☐ Partially: The paper briefly describes the operation sequences.
- ☐ No: The paper does not include any operation representation.

QQ10: Is a code repository available?

- ☐ Yes: The authors report a code repository with the environment/dataset used.
- ☐ Partially: The authors reported only the environment or dataset used.
- ☐ No: The authors do not report any piece of code.

After applying the quality criteria, the results in Tables 26 and 27 were obtained

A.4 Extracted data

From each article, the following information was be extracted:

- Title: Original title of the publication.
- Authors: Name of all authors of the article.
- Country: Location of the main author's affiliation.
- Publication date: Date in which the article was first published.

Table 26 Individual results for quality criteria, part 1

Article	QQ01	QQ02	QQ03	QQ04	QQ05	QQ06	QQ07	QQ08	QQ09	QQ10	Score
Alarcon et al. (2021)	Y	Y	Y	Y	P	Y	Y	P	Y	N	8.0
Zhao et al. (2021b)	P	P	Y	Y	Y	P	Y	Y	Y	N	7.5
Abdulov et al. (2022)	Y	Y	Y	Y	P	N	Y	Y	Y	N	7.5
Peake et al. (2020)	Y	Y	P	P	P	N	Y	Y	N	Y	6.5
Chen and Lu (2022)	N	P	Y	N	Y	Y	Y	P	Y	P	6.5
Grando et al. (2020)	P	Y	Y	N	P	Y	P	Y	N	Y	6.5
Dang et al. (2018)	Y	Y	Y	P	P	N	P	P	Y	N	6.0
You et al. (2020)	N	P	P	N	Y	Y	Y	Y	N	Y	6.0
Li and Hu (2021)	N	P	N	Y	P	Y	Y	Y	Y	N	6.0
Zhao et al. (2021a)	P	Y	Y	N	P	N	Y	Y	Y	N	6.0
Mumuni et al. (2022)	N	P	N	N	Y	Y	Y	Y	Y	N	5.5
Hu et al. (2022)	P	P	Y	N	P	N	Y	Y	Y	N	5.5
Mansouri et al. (2020b)	N	Y	P	Y	P	P	Y	P	P	N	5.5
Fu et al. (2022)	N	P	Y	N	P	Y	Y	Y	N	N	5.0
Delamer et al. (2021)	P	P	Y	N	P	P	P	P	Y	N	5.0
Bachrach et al. (2010)	P	Y	P	Y	P	N	P	P	N	N	4.5
Yousuf et al. (2022)	Y	P	P	N	P	N	P	P	Y	N	4.5
Capi et al. (2021)	P	Y	N	Y	N	N	P	Y	P	N	4.5
Dang et al. (2020)	Y	Y	N	Y	P	N	P	P	N	N	4.5
Nakamura et al. (2017)	N	Y	Y	P	P	N	P	N	P	N	4.0
Mao et al. (2020)	P	P	P	N	P	N	P	P	Y	N	4.0
Zhang et al. (2018)	P	P	N	N	P	P	P	P	Y	N	4.0
Mansouri et al. (2020a)	P	Y	N	Y	P	N	Y	N	N	N	4.0
Bershadsky and Johnson (2013)	Y	P	P	N	N	N	P	P	Y	N	4.0
Jia et al. (2022)	N	P	N	N	P	P	P	Y	Y	N	4.0

Table 27 Individual results for quality criteria, part 2

Article	QQ01	QQ02	QQ03	QQ04	QQ05	QQ06	QQ07	QQ08	QQ09	QQ10	Score
Kusnur et al. (2021)	P	P	P	N	P	N	N	P	Y	N	3.5
Fuentes et al. (2014)	P	P	N	Y	P	N	P	P	N	N	3.5
Cho et al. (2022)	N	P	Y	Y	P	N	P	N	N	N	3.5
Nitsche et al. (2020)	N	P	Y	Y	P	N	P	N	N	N	3.5
Nakamura et al. (2016)	N	P	Y	Y	P	N	P	N	N	N	3.5
Helgesen et al. (2019)	N	P	P	P	P	P	P	N	N	N	3.0
Prophet and Trommer (2020)	P	Y	P	N	P	N	P	N	N	N	3.0
Albanese et al. (2022)	N	N	N	P	P	N	Y	Y	N	N	3.0
Bhusal et al. (2022)	N	N	N	P	Y	N	P	Y	N	N	3.0
Fu et al. (2013)	N	P	N	P	P	P	N	P	P	N	3.0
Perez-Grau et al. (2021)	P	P	N	Y	P	N	P	N	N	N	3.0
Cao (2015)	N	Y	P	N	P	N	Y	N	N	N	3.0
Nagarani et al. (2020)	N	P	P	N	P	Y	P	N	N	N	3.0
Cocchioni et al. (2014)	P	P	N	Y	P	N	P	N	N	N	3.0
Al-Jarrah et al. (2013)	P	P	N	P	P	N	P	P	N	N	3.0
Narazaki et al. (2022)	N	P	Y	N	N	N	P	N	Y	N	3.0
Kumar et al. (2020)	P	P	P	P	P	N	N	N	P	N	3.0
Wang et al. (2017)	P	P	P	P	P	N	P	N	N	N	3.0
Xu et al. (2013)	N	P	P	P	P	N	P	N	N	N	2.5
Yang et al. (2021)	N	P	N	Y	P	N	P	N	N	N	2.5
Louali et al. (2017)	P	P	P	N	P	N	P	N	N	N	2.5
Li et al. (2018)	N	P	N	Y	P	N	P	N	N	N	2.5
Yu et al. (2017)	N	P	Y	N	P	P	N	N	N	N	2.5

Table 27 continued

Article	QQ01	QQ02	QQ03	QQ04	QQ05	QQ06	QQ07	QQ08	QQ09	QQ10	Score
Tsiourva and Papachristos (2020)	N	P	N	Y	P	N	P	N	N	N	2.5
Sun et al. (2019)	N	P	P	N	P	N	Y	N	N	N	2.5
Lugo-Cárdenas et al. (2017)	N	P	Y	N	P	N	P	N	N	N	2.5
Zhang and Duan (2014)	N	P	N	N	P	N	P	N	Y	N	2.5
Zhang et al. (2017)	N	P	P	P	P	N	P	N	N	N	2.5
Meng et al. (2019)	N	P	P	N	P	N	P	N	N	N	2.0
Song et al. (2022)	N	P	N	Y	N	N	P	N	N	N	2.0
He et al. (2017)	N	P	N	N	P	P	P	N	N	N	2.0
Tymochko et al. (2020)	N	P	N	N	P	N	P	N	N	N	2.0
Konrad et al. (2017)	N	P	N	P	P	N	P	P	N	N	2.0
Zaki et al. (2017)	N	P	N	N	P	P	N	P	N	N	2.0
Parfiryev et al. (2020)	P	N	N	N	P	N	P	N	N	N	1.5
Lee and Kim (2022)	N	P	N	N	P	N	P	N	N	N	1.5
de Plinval et al. (2017)	N	P	N	N	P	N	P	N	N	N	1.5
Zhu et al. (2018)	N	P	N	N	P	N	P	N	N	N	1.5
Sarmiento and Murphy (2018)	N	P	N	N	P	N	P	N	N	N	1.5
Michaelsen et al. (2012)	N	P	P	N	N	N	N	N	N	N	1.0
Michaelsen et al. (2011)	N	P	P	N	N	N	N	N	N	N	1.0
Maitra et al. (2016)	N	P	N	N	N	N	N	N	N	N	0.5

- Publication venue: Name of the journal, magazine, conference, or any other source where the article was originally published.
- Problem that is solved: Articles' problem formulation.
- Methodology summary: Main methods presented or used by the authors, categorized to show which navigation tasks they accomplish.
- UAV type: Category of flying vehicle addressed by the solution.
 - Fixed wing
 - Helicopter
 - Quadcopter
 - Hexacopter
 - Multicopter
 - Blimp
- Scenario(s) type: Category of scenario in which the method was implemented.
 - Equation space
 - Indoor
 - Outdoor
- Sensory data used: UAV sensor data acquired to work with the article's methods.
- Simulation software: Software used for testing the method in simulation.
- Benchmark metrics: Metrics used to compare the method against others.
- Presents a real-world application?: Check if the method was tested on real-world scenarios.
 - Yes
 - No
- Real-world scenario specs: Specifications of the real-world setup.
- Application domain: Which tasks were solved by the method.
 - Aerial refueling
 - Competition
 - Control
 - Exploration
 - Landing
 - Localization
 - Mapping
 - Obstacle detection
 - Path planning
 - Predefined navigation
 - Target recognition
- Environment application: Type of environment where the method was tested.
 - Arbitrary
 - Dynamic
 - Known
 - Landmarks
 - Pathway
 - Static
 - Structured

- Control state variables: Values used to achieve UAV flight control.
- Goal specification: The type of goal that the vehicle must reach.
 - Object goal
 - Point goal
 - Unspecified
- Future studies: The future steps reported by the authors.

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Declarations

Competing interests The authors declare no competing interests.

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